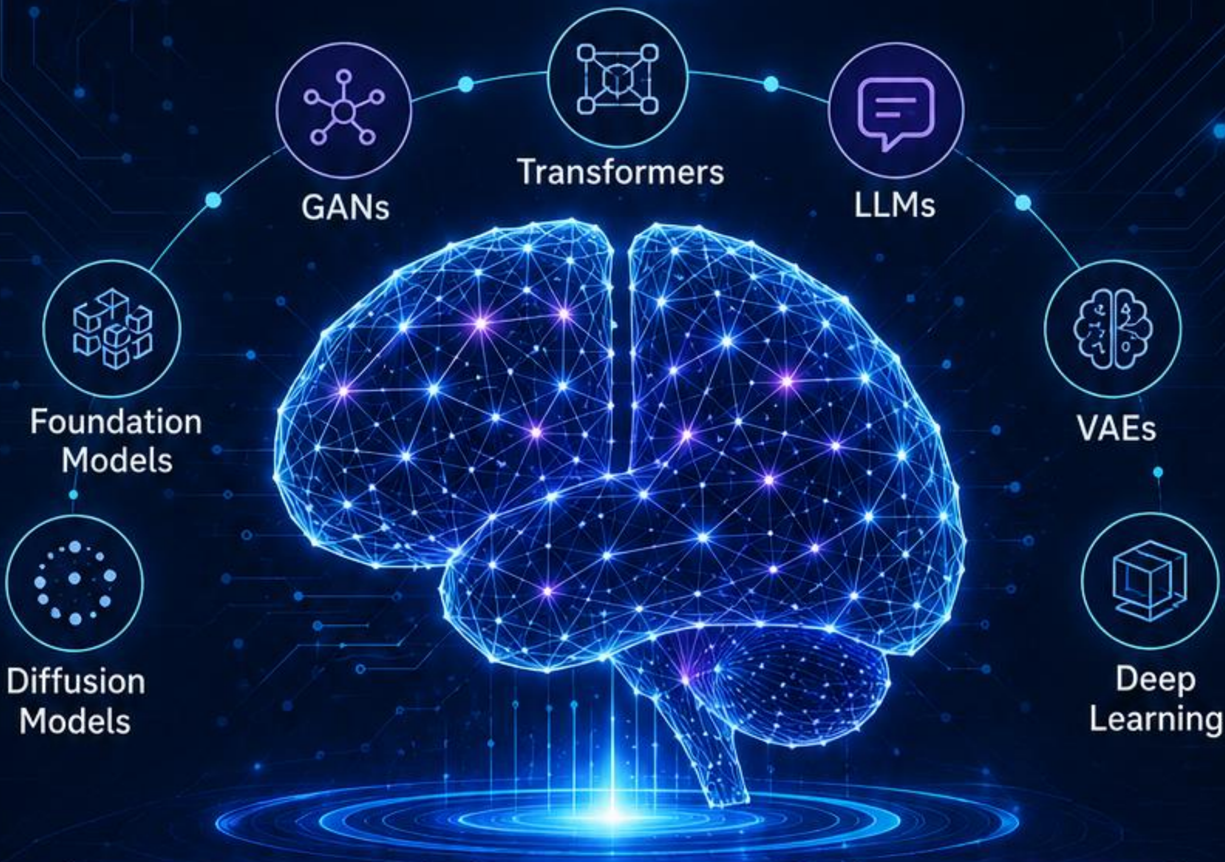
GENERATIVE AI

Foundation Models & Platforms

Comprehensive Learning Notes



IBM Granite

GPT
OpenAI

Llama
Meta

watsonx

Hugging Face

TOPICS COVERED

- Foundation Models
- LLMs
- Transformers
- GANs
- VAEs
- Diffusion Models

PREPARED BY

KASHIF MAQBOOL JOIYA

Data Scientist | AI Engineer

github.com/KashifMaqbool

Table of Content

Module 1: Models for Generative AI

1 IBM Product Spotlight: watsonx.governance

1.1 Introduction

IBM **watsonx.governance** is a platform designed to manage the entire AI model lifecycle, from development through production, while providing built-in compliance and risk management capabilities. It helps organizations develop, deploy, and maintain AI models responsibly by ensuring transparency, fairness, and regulatory compliance.

1.1.1 Risk Management in the AI Lifecycle

- Risk management is the proactive process of **identifying, assessing, and mitigating potential negative outcomes** associated with developing and deploying AI.
- Unlike traditional software, AI models are dynamic, probabilistic, and can behave unpredictably when exposed to new data.

1.1.2 Compliance in the AI Lifecycle

- Compliance refers to ensuring that an AI model adheres to **external laws, government regulations, and internal organizational policies**.
- Because AI can impact privacy, civil rights, and safety, compliance acts as the guardrails that keep an organization legally protected.

1.2 Key Capabilities of IBM watsonx.governance

- Monitor AI model performance and fairness.
- Manage the AI model lifecycle from development to production.
- Ensure compliance with AI regulations and governance requirements.
- Build trust in AI through transparency and explainability (the ability to understand and interpret how an AI model makes its decisions).

1.3 Learn More

IBM watsonx.governance: http://ibm.biz/genaiprod-wxgov

2 Generative AI: Introduction and Applications – Overview

2.1 Introduction

Generative AI is a rapidly developing field that is changing how people create content, solve problems, and build intelligent applications. This overview introduces the core ideas, models, and platforms that shape the field and provides a practical starting point for understanding how generative systems work.

The focus is on the fundamental concepts and model families that form the foundation of generative AI. These include deep learning and Large Language Models (LLMs), along with major model types such as:

- Generative Adversarial Networks (GANs)
- Variational Autoencoders (VAEs)
- Transformers
- Diffusion Models

The overview also introduces the idea of Foundation Models and explains how pre-trained models can be used to generate:

- Text
- Images
- Code

In addition, it highlights widely used generative AI platforms, including:

- IBM watsonx
- Hugging Face

Practical labs provide hands-on exposure through the IBM Generative AI Classroom and platforms such as IBM watsonx. These activities include working with foundation models such as:

- IBM Granite
- OpenAI GPT
- Google FLAN
- Meta Llama

Expert insights are also included to show how generative AI is being applied in real-world settings and what tools are shaping its growth.

2.2 Who This Is For

This overview is suitable for anyone interested in the rapidly evolving field of Generative AI and its core concepts, models, and platforms.

It is especially useful for:

- Individuals seeking an introduction to Generative AI models, platforms, and their capabilities.
- Professionals looking to improve their work by leveraging the power of Generative AI.
- Managers and executives exploring how Generative AI platforms can be used within their organizations.
- Students who want to build practical Generative AI skills to improve their job readiness.

2.3 Recommended Background

This material is designed for anyone interested in exploring Generative AI and requires **no specific prerequisites**.

It uses simple, easy-to-understand language to explain the essential concepts without relying on technical jargon.

The hands-on labs require **no programming experience**, and **no educational degree is required**.

To get the most from the material, active participation and completion of the learning activities across all sections are recommended.

3 Deep Learning and Large Language Models

3.1 Introduction

In this topic, the core concepts of **Generative AI**, **Deep Learning**, and **Large Language Models (LLMs)** are explained. It also describes how LLMs perform human-like tasks by leveraging deep learning techniques and transformer-based architectures.

3.2 Deep Learning

Deep learning is inspired by the way the human brain processes information. Just as humans develop a deeper understanding by processing information through multiple layers, deep learning creates "depth" by processing data through multiple computational layers.

Deep learning is made possible through an **Artificial Neural Network (ANN)**.

3.2.1 Artificial Neural Networks (ANNs)

An Artificial Neural Network (ANN) consists of several computing units called **neurons**, which are organized into three connected layers:

- **Input Layer** – Captures the input data.
- **Hidden Layer(s)** – Processes and learns patterns from the data. There can be one or many hidden layers.
- **Output Layer** – Produces the final prediction or output.

When a vast dataset is introduced into the network:

- Neurons in the input layer capture the data.
- Neurons in the hidden layers study and process the data.
- The output layer generates the final result.

3.2.2 Parameters in Neural Networks

Each neuron in the hidden layer contains **bias parameters**, while the connections between neurons contain **weight parameters**.

Parameters are internal values of a neural network that are optimized as the neurons repeatedly train on large datasets.

The greater the number of **bias** and **weight** parameters:

- The stronger the computational power of the network.
- The higher the predictive accuracy.

3.3 Types of Deep Learning Training

Deep learning algorithms are commonly trained using either **supervised learning** or **unsupervised learning**.

3.3.1 Supervised Learning

In supervised learning, algorithms are trained on **labeled datasets**, where every input has a known output.

Applications include:

- Email filtering
- Credit score evaluation
- Fraud detection
- Image recognition
- Voice recognition

3.3.1.1 Limitations of Supervised Learning

Although better labeling can improve algorithm quality, supervised learning has several limitations:

- Algorithms are restricted to predefined responses.
- Creating labeled datasets is time-consuming.
- Labeling data is expensive.

3.3.2 Unsupervised Learning

More commonly, deep learning algorithms are trained on **unlabeled datasets**, where the training data contains inputs without explicit target outputs.

Common applications include:

- Clustering
- Dimensionality Reduction

3.3.2.1 Clustering

Clustering groups similar data instances together based on their inherent properties.

3.3.2.2 Dimensionality Reduction

Dimensionality reduction captures the most important features of data while discarding redundant or less informative information.

Because of this, unsupervised learning algorithms are better able to discover patterns and hierarchies within datasets, often producing more efficient and accurate results.

3.4 Factors Affecting Deep Learning Performance

The quality of responses generated by deep learning algorithms depends mainly on:

- The quality of the large dataset used for training.
- The neural network architecture employed.

3.5 Deep Learning Architectures

Three major deep learning architectures are commonly used:

- Convolutional Neural Networks (CNNs)
- Recurrent Neural Networks (RNNs)
- Transformer-Based Models

3.5.1 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) consist of multiple layers, each performing a **convolution** (a mathematical operation) on the output of the previous layer.

When applied to grid-based data such as images, CNNs can efficiently:

- Extract useful information from images.
- Recognize patterns.
- Classify images.
- Segment images.

Applications of CNNs include:

- Image processing
- Video recognition
- Natural Language Processing (NLP)

3.5.2 Recurrent Neural Networks (RNNs)

Recurrent Neural Networks (RNNs) are designed for processing **sequential data**, such as text and speech.

A key characteristic of RNNs is their **memory component**, which enables them to capture dependencies and contextual information over time.

Applications of RNNs include:

- Machine translation
- Sentiment analysis
- Speech recognition

3.5.3 Transformer-Based Models

Transformer-based models do not rely on convolutions or recurrence for processing data.

Instead, they use a **two-stack architecture** consisting of:

- Encoder
- Decoder

These components process an exceptionally large number of parameters to understand language patterns more effectively.

Transformer models can:

- Analyze the context of words.
- Understand relationships between words.
- Capture hierarchical language patterns.
- Predict the next word in a sequence.

This enables the creation of **Large Language Models (LLMs)** capable of performing various **Natural Language Processing (NLP)** tasks.

Applications include:

- Content generation
- Predictive analysis
- Language translation
- Process automation

3.6 Large Language Models (LLMs)

Large Language Models (LLMs) serve as the foundational mechanism behind Generative AI applications.

Examples of LLMs include:

- OpenAI's Generative Pre-trained Transformers (GPT-3 and GPT-4)
- Google's PaLM 2
- Meta's Llama

3.6.1 GPT-4 Example

GPT-4 is a language processing AI model trained on a massive corpus of Internet text, including:

- Books
- Articles
- Websites

The model contains **over 170 trillion parameters**, enabling it to perform advanced Natural Language Processing tasks such as:

- Content creation
- Dialogue systems
- Language translation

People use GPT-4 to:

- Write high-quality essays.
- Prepare case papers.
- Perform machine translation.
- Generate summaries.

Organizations use GPT-4 to:

- Power chatbots.
- Build virtual agents.
- Translate international business communications.
- Localize website content into different languages.

3.7 Future of Large Language Models

As deep learning architectures and technologies continue to evolve, Large Language Models will become increasingly capable of delivering more accurate and reliable outcomes, enabling Generative AI systems to perform increasingly complex tasks.

3.8 Key Takeaways

- Deep learning is inspired by the layered information-processing approach of the human brain.
- Artificial Neural Networks (ANNs) enable deep learning through interconnected neurons organized into input, hidden, and output layers.
- Neural networks learn by optimizing **bias** and **weight** parameters.
- Deep learning models can be trained using supervised or unsupervised learning.

- The quality of training data and the neural network architecture significantly influence model performance.
- The three primary deep learning architectures are CNNs, RNNs, and Transformer-based models.
- Transformer networks form the foundation of Large Language Models (LLMs).
- LLMs leverage transformer architectures to pretrain deep learning algorithms on vast datasets.
- LLMs capture patterns and hierarchies within data to generate accurate, human-like responses.
- This capability makes Generative AI scalable and enables applications such as content generation, translation, predictive analysis, and intelligent automation.

4 Generative AI Models

4.1 Introduction to Generative AI Models

In the world of Generative AI, four core models have made a significant impact:

- Variational Autoencoders (VAEs)
- Generative Adversarial Networks (GANs)
- Transformer-Based Models
- Diffusion Models

Each model employs a different type of deep learning architecture and applies probabilistic techniques (methods based on probability to learn patterns from data).

4.2 Variational Autoencoders (VAEs)

Variational Autoencoders (VAEs) are among the most popular generative AI models for two main reasons:

- They work with a diverse range of training data, such as images, text, and audio.
- They rapidly reduce the dimensionality (reduce the number of variables while preserving important information) of images, text, or audio to create a newer, improved version.

4.2.1 How VAEs Work

VAEs consist of two self-sufficient neural networks:

- **Encoder**
- **Decoder (Reverse Encoder)**

The process works as follows:

- The encoder studies the probability distribution (how the data values are statistically distributed) of the input data.
- In simple terms, it isolates the most useful data variables.

- The encoder creates a compressed representation of the data sample and stores it in the **latent space** (a mathematical space within the model where high-dimensional data is represented in a compressed format).
- The decoder then decompresses the compressed representation stored in the latent space to generate the desired output.

4.2.2 Training Principle

VAEs are trained using the **Maximum Likelihood Principle** (an optimization method that minimizes the difference between the original input data and the reconstructed output).

Although VAEs are trained in a static environment, their latent space is continuous. Therefore, they can generate new samples by randomly sampling from the probability distribution of the data.

4.2.3 Applications of VAEs

Because VAEs can produce realistic and varied images with relatively little training data, they are commonly used for:

- Image synthesis
- Data compression
- Anomaly detection

4.2.3.1 Industry Applications

Entertainment Industry

- Creating game maps
- Animating avatars

Finance Industry

- Forecasting the volatility surfaces of stocks (predicting how stock price fluctuations may change over time)

Healthcare Sector

- Detecting diseases using electrocardiogram (ECG) signals

4.3 Generative Adversarial Networks (GANs)

Generative Adversarial Networks (GANs) are another type of generative AI model that works with imagery and textual input data.

4.3.1 How GANs Work

GANs consist of two **Convolutional Neural Networks (CNNs)** that compete against each other in an adversarial game.

The two networks are:

- **Generator**
    - Trained on a vast dataset.
    - Produces new data samples.
- **Discriminator**
    - Determines whether a sample is real or fake.
    - Provides feedback to the generator.

Based on the discriminator's responses, the generator continually improves and produces increasingly realistic data samples.

4.3.2 Capabilities of GANs

GANs can:

- Generate realistic-looking images.
- Perform style transfer.
- Perform image-to-image translation.
- Create deepfakes.

4.3.3 Industry Applications

Finance Industry

- Training models for loan pricing
- Generating time-series data

Other Applications

- **SpaceGAN** works with geospatial data and videos.
- **StyleGAN2** is known for creating video game characters.

4.3.4 Limitations of GANs

Compared to VAEs, GANs:

- Require a large amount of training data.
- Need significant computational power.
- Can potentially create false or misleading material, raising ethical concerns.

4.4 Transformer-Based Models

Transformer-based models were introduced when **Recurrent Neural Networks (RNNs)** began experiencing the **vanishing gradient problem** (a problem where gradients become extremely small during training, making it difficult for the network to learn long-term dependencies).

Because of this issue, RNNs struggled to process long sequences of text effectively.

4.4.1 How Transformers Solve This Problem

Transformers use **attention mechanisms** (techniques that allow the model to focus on the most relevant parts of the input while ignoring less important information).

This enables transformers to:

- Focus on the most valuable parts of the text.
- Filter out unnecessary information.
- Model long-term dependencies in text.

4.4.2 Transformer Architecture

When a prompt is entered, the **two-stack transformer architecture** uses an encoder-decoder mechanism to generate coherent and contextually relevant text.

4.4.3 Capabilities of Transformers

Transformer models can query extensive databases, allowing them to:

- Build Large Language Models (LLMs)
- Perform Natural Language Processing (NLP) tasks
- Create images
- Synthesize music
- Generate videos

This represents a major breakthrough in content creation and has enabled innovations such as:

- GPT-3.5 and its later versions
- BERT
- T5

4.5 Diffusion Models

Diffusion models are a more recent addition to the world of Generative AI.

They address the systematic decay of data that occurs due to noise in the latent space.

By applying the principles of diffusion, these models aim to prevent information loss.

4.5.1 How Diffusion Models Work

Similar to the diffusion process, where molecules move from high-density areas to low-density areas, diffusion models move noise to and from a data sample through a two-step process.

4.5.1.1 Step 1: Forward Diffusion

- Algorithms gradually add random noise to the training data.

4.5.1.2 Step 2: Reverse Diffusion

- Algorithms reverse the noise-adding process.
- They recover the original data.
- They generate the desired output.

4.5.2 Popular Diffusion Models

Some mature diffusion models include:

- OpenAI's DALL·E 2
- Stability AI's Stable Diffusion XL
- Google's Imagen

These models generate high-quality graphical content.

4.5.3 Relationship with VAEs

Like Variational Autoencoders, diffusion models also optimize data by:

- Projecting it onto the latent space.
- Recovering it back to its initial state.

However, diffusion models are trained using a dynamic flow, which makes training take longer.

4.5.4 Why Diffusion Models Are Highly Effective

Diffusion models are considered among the best options for generative AI because they:

- Train hundreds, or potentially an unlimited number, of layers.
- Produce remarkable results in image synthesis.
- Produce remarkable results in video generation.

4.6 Ongoing Research

Experiments with generative AI models continue unabated as unsupervised algorithms continue to produce new and surprising advancements.

4.7 Key Takeaways

- **Variational Autoencoders (VAEs)** rapidly reduce the dimensionality of samples and reconstruct them through latent space representations.
- **Generative Adversarial Networks (GANs)** use competing generator and discriminator networks to produce realistic samples.
- **Transformer-Based Models** use attention mechanisms to model long-term dependencies and power modern Large Language Models.
- **Diffusion Models** generate high-quality content by gradually removing noise from latent representations and are particularly effective for image and video generation.

5 Foundation Models

5.1 What Are Foundation Models?

The Stanford University Center for Research on Foundation Models defines a **foundation model** as:

"A new successful paradigm for building AI systems: Train one model on a huge amount of data and adapt it to many applications. We call such a model a foundation model."

This definition highlights two important concepts:

- Train one model on a huge amount of data.
- Adapt the trained model to many different applications.

5.2 How Foundation Models Work

A foundation model is a **large, general-purpose, self-supervised model** that is pre-trained on vast amounts of **unlabeled data**, establishing **billions of parameters** (internal values the model learns during training to recognize patterns and make predictions).

5.2.1 Pre-training

Pre-training is a technique in which unsupervised algorithms are repeatedly allowed to make connections between diverse pieces of information.

This enables foundation models to develop:

- **Multimodal capabilities** (the ability to understand and generate multiple data types such as text, images, audio, and video).
- **Multi-domain capabilities** (the ability to work across different fields and applications).

5.3 Capabilities of Foundation Models

Because of their broad pre-training, foundation models can accept prompts in multiple formats, including:

- Text
- Images
- Audio
- Video

They can perform numerous complex and creative tasks, including:

- Answering questions
- Summarizing documents
- Writing essays
- Solving equations

- Extracting information from images
- Developing code

This broad skill set makes foundation models useful across many different domains.

5.4 Foundation Models vs. Smaller Generative AI Models

Foundation models differ from smaller generative AI models in several ways.

Foundation Models

- Trained on massive amounts of diverse, unlabeled data.
- Perform a wide variety of tasks.
- Work across multiple domains.
- Can be adapted to many applications.

Smaller Generative AI Models

- Trained on restricted domain-specific datasets.
- Designed for limited tasks.
- Have narrower capabilities.

5.4.1 Examples

Foundation Model

- OpenAI's DALL·E family of models is considered a foundation model because it can perform many image-related tasks.

Not a Foundation Model

- AlexNet is **not** classified as a foundation model because it only performs image classification tasks.

5.4.2 Important Clarification

- All foundation models have generative AI capabilities.
- Not all generative AI models are foundation models.

5.5 Large Language Models (LLMs)

When foundation models are trained on vast Natural Language Processing (NLP) datasets, they are called **Large Language Models (LLMs)**.

LLMs develop **independent reasoning**, allowing them to respond to queries in unique ways.

5.5.1 Examples of Large Language Models

- **OpenAI GPT Family**

    - GPT-3: Pre-trained on more than **175 billion parameters**.
    - GPT-4: Estimated to be pre-trained on more than **180 trillion parameters**.
- **Google PaLM (Pathways Language Model)**
    - Pre-trained on **540 billion parameters**.
- **Meta LLaMA (Large Language Model Meta AI)**
    - Pre-trained on **65 billion parameters**.
- **Google BERT**
    - Pre-trained on more than **340 million parameters**.
- **Meta Galactica**
    - Designed for scientists.
    - Pre-trained on **48 million papers, lectures, textbooks, and websites**.
- **Falcon 7B (Technology Innovation Institute)**
    - Pre-trained on **1.5 trillion tokens**.
- **Microsoft Orca**
    - Pre-trained on **13 billion parameters**.
    - Small enough to run on a laptop.

Note: These parameter counts may change as generative AI models continue to evolve.

5.6 Adaptability of Foundation Models

A key characteristic of foundation models is their ability to adapt to many applications.

Because they receive broad-based training, foundation models can:

- Learn new tasks.
- Adapt to new situations.
- Be fine-tuned for specific applications.

This enables small businesses to create customized and efficient generative AI models at a much lower cost.

For this reason, foundation models are also called **base models**.

They make AI systems more accessible to businesses and individuals who lack the resources to train models from scratch.

As a result, foundation models enable enterprises to reduce **time-to-value** (the time required to gain practical business benefits) from months to weeks.

5.7 Foundation Models in Chatbots

The evolution of chatbots demonstrates the power of foundation models.

5.7.1 Modern Chatbots

- OpenAI GPT-3 and GPT-4 power ChatGPT.
- Google PaLM powers Google's Bard chatbot.

These chatbots are significantly more capable than earlier chatbot systems.

5.7.2 Earlier Chatbots

Earlier chatbots:

- Were trained on much smaller datasets.
- Predicted responses using keywords.
- Produced predetermined responses.
- Had limited generative capabilities.

5.7.3 Today's Chatbots

Modern chatbots:

- Are pre-trained multiple times on extensive datasets.
- Have higher word prediction accuracy.
- Produce more helpful, creative, and context-aware responses.

For example, a single prompt may allow ChatGPT to:

- Write a comparative essay.
- Create an infographic.
- Design a checklist.
- Write a short story.

5.8 Foundation Models for Image Generation

Foundation models are also used for image generation.

5.8.1 DALL·E

OpenAI's GPT-3 serves as the foundation model for **DALL·E**, an image generation tool that responds to text prompts.

For a single text prompt, DALL·E can generate:

- Four high-resolution images.
- Photorealistic images.
- Paintings.
- Multiple artistic styles.

5.8.2 Another Important Clarification

- All Large Language Models are foundation models.
- Not all foundation models are Large Language Models.

Some foundation models specialize in image generation rather than language understanding.

5.9 Diffusion-Based Foundation Models

Some foundation models use **diffusion architecture** to improve image generation capabilities.

5.9.1 Examples

DALL·E

- Uses transformer architecture.
- The latest version incorporates diffusion techniques to generate images from text.

Stable Diffusion

- Uses diffusion architecture.
- Generates high-resolution images.
- Produces realistic, cartoon, and abstract styles based on user descriptions.

Google Imagen

- Uses a cascaded diffusion model built on a Large Language Model.
- Generates images from text prompts.

5.10 Limitations of Foundation Models

Although foundation models are highly capable, they also have limitations.

5.10.1 Bias

If the training data is biased, the generated output may also be biased.

5.10.2 Hallucinations

Large Language Models may **hallucinate** (generate false or inaccurate information because they incorrectly interpret relationships within the learned data).

Therefore, it is important to verify the accuracy of responses generated by AI chatbots.

With careful use, foundation models provide significant benefits despite these limitations.

5.11 Key Takeaways

- Foundation models are large, self-supervised models pre-trained on vast amounts of unlabeled data.
- They are trained on billions of parameters, enabling them to perform a wide variety of complex tasks.
- Their multimodal and multi-domain capabilities allow them to serve as the foundation (or base) for numerous generative AI applications.

- Large Language Models (LLMs) are foundation models trained primarily on natural language data.
- Foundation models can be adapted for many applications, reducing the cost and time required to build AI systems.
- Despite their capabilities, foundation models can produce biased outputs and hallucinated information, making human verification essential.

Generative AI Model	Typical Deep Learning Architecture(s)	Explanation
Variational Autoencoders (VAEs)	CNNs, RNNs, Transformers	VAEs are an encoder–decoder framework. The encoder and decoder can be implemented using CNNs (images), RNNs (sequences), or Transformers (text and multimodal data).
Generative Adversarial Networks (GANs)	CNNs, RNNs, Transformers	GANs consist of a Generator and a Discriminator. For image generation, both are typically CNNs (e.g., DCGAN, StyleGAN). For sequential data, RNNs may be used. Some modern GANs also use Transformers.
Transformer-Based Models	Transformers	The model itself is based on the Transformer architecture. Examples include GPT, BERT, T5, and LLaMA.
Diffusion Models	CNNs (U-Net), Transformers, Hybrid Architectures	Early diffusion models (e.g., DDPM, Stable Diffusion) primarily use a CNN-based U-Net. Newer models, such as DiT (Diffusion Transformers), replace the U-Net with a Transformer.

6 GPT-5 and Google Gemini: Multimodal Foundation Models

6.1 Introduction

In the dynamic landscape of Artificial Intelligence (AI), an enthralling competition unfolds between two leading foundation models: **OpenAI's GPT-5** and **Google's Gemini**. Both models showcase remarkable capabilities, pushing the limits of AI potential. However, they also forge distinct paths with unique strengths and approaches.

This topic explores the features and nuances that distinguish these two AI foundation models.

6.2 GPT-5

GPT-5 is the latest Large Language Model (LLM) developed by OpenAI. It is an advanced AI capable of:

- Understanding and generating human-like text.
- Answering questions.
- Writing code.

- Summarizing content.
- Assisting with reasoning tasks.

It improves upon earlier versions, such as GPT-4, with better accuracy, improved context understanding, and enhanced multimodal abilities (working with text, images, and more).

6.2.1 Key Strengths of GPT-5

- **Advanced Comprehension and Reasoning**
    - GPT-5 can understand complex instructions.
    - It maintains context over long conversations.
    - It reasons through multi-step problems more effectively than earlier versions.
    - This makes it highly useful for research, coding, and academic tasks requiring logical depth.
- **Multi-Modal Intelligence**
    - GPT-5 can analyze and generate not only text but also work with images, charts, and code.
    - It can interpret diagrams, debug visual data, and create data-driven visualizations within a single workflow.
- **Higher Accuracy and Context Awareness**
    - GPT-5 produces more precise and relevant responses by reducing factual errors.
    - It remains consistent with the user's intent.
    - Its improved contextual memory (ability to remember previous parts of a conversation) helps maintain coherence across long interactions.

6.3 Google Gemini

Google Gemini is a multimodal foundation model capable of accomplishing intricate tasks across diverse domains such as mathematics, physics, and beyond. It also demonstrates an exceptional ability to understand and generate high-quality code in various programming languages.

6.3.1 Key Strengths of Google Gemini

- **Handling Reasoning and Mathematics**
    - Compared to GPT-5, Gemini demonstrates stronger logic and deduction skills.
    - It excels at intricate mathematical problems.
    - It engages in complex, reasoned arguments.
    - These capabilities make it particularly suitable for advanced scientific and technological applications where GPT-5 may still face occasional limitations with highly complex multi-step reasoning.
- **Extending Beyond Text**
    - Gemini extends its capabilities beyond text by handling audio, images, and video seamlessly.
    - It can analyze visuals.
    - Generate lifelike images.
    - Compose soundtracks.
    - This makes it well-suited for multimedia projects where GPT-5 primarily focuses on text, images, and code.

- **On-Device Processing**
  - Gemini offers on-device processing (running AI directly on the user's device instead of remote cloud servers).
  - This enhances privacy by keeping data on the device.
  - It can also reduce latency (delay in processing), providing a faster user experience.

6.4 Comparison Between GPT-5 and Google Gemini

6.4.1 Text Generation

GPT-5

- Captures writing style and nuance exceptionally well.
- Produces natural and engaging text across diverse genres.
- Can generate:
  - Mystery novels.
  - Documentary scripts.
  - Creative stories.
  - Professional content.

Gemini

- Excels at producing informative and factual content.
- Particularly effective for:
  - Research summaries.
  - Scientific reports.
  - Technical documentation.

6.4.2 Reasoning and Problem-Solving

GPT-5

- Excels at both basic and advanced reasoning.
- Effectively analyzes complex historical narratives.
- Performs well on multi-step logical reasoning tasks.
- Extremely intricate puzzles, such as Sudoku, may still present occasional challenges.

Gemini

- Equipped with robust reasoning modules.
- Excels in logic and deduction.
- Can:
  - Solve intricate mathematical problems.
  - Identify patterns in large datasets.
  - Engage in coherent reasoned debates.
  - Analyze scientific data to propose new hypotheses.
  - Formulate strategies for complex games.

6.4.3 Multimodality

GPT-5

- Works seamlessly with:
  - Text.
  - Images.
  - Code.
- Can:
  - Interpret visual content.
  - Generate detailed captions.
  - Analyze image-based data.
  - Produce functional code snippets.
- Tasks involving ultra-realistic image generation or extensive software development may extend beyond its primary scope.

Gemini

- A truly multimodal model capable of interacting with:
  - Text.
  - Images.
  - Audio.
  - Video.
- It can:
  - Analyze video footage.
  - Generate realistic images.
  - Compose soundtracks.
  - Analyze medical scans.
  - Create artwork based on emotions.
  - Produce movie trailers with original background music.

6.4.4 Accessibility and Privacy

GPT-5

- Primarily accessed through cloud-based platforms.
- Provides:
  - Scalable performance.
  - Easy updates.
- Users should consider:
  - Data privacy.
  - Possible network latency.

Gemini

- Supports both:
  - Cloud processing.
  - On-device processing.
- This provides flexibility between convenience and enhanced privacy.

- It enables sensitive information to remain on the user's device without relying on external servers.

6.4.5 Customization and Fine-Tuning

GPT-5

- Offers improved customization compared to earlier versions.
- Users can adjust:
  - Prompts.
  - Tone.
  - Writing style.
- Customization remains within predefined safety and reliability constraints.

Gemini

- Expected to provide a higher level of customization and fine-tuning.
- Can potentially adapt to:
  - Individual writing styles.
  - Specific user preferences.
  - Specialized reasoning tasks.
  - Domain-specific problem-solving.

6.5 Key Takeaways

- **GPT-5** is a highly capable multimodal foundation model that excels in natural language understanding, reasoning, coding, contextual awareness, and high-quality text generation.
- **Google Gemini** is a multimodal foundation model with strengths in advanced reasoning, mathematics, multimedia processing, privacy through on-device execution, and extensive customization.
- GPT-5 is generally stronger in natural language generation and maintaining conversational context.
- Gemini distinguishes itself through broader multimodal capabilities, stronger mathematical reasoning, and flexible deployment options, including on-device processing.

7 Generative AI Foundation Model Testing

7.1 Exercise 1: Classify Text and Detect Sentiment Using a Foundation Model

7.1.1 Objective

Learn how to use the **Gemini 2.5 Flash** foundation model to classify text and identify sentiment.

7.1.2 Set Up the AI Environment

1. Create or rename your chat as needed for the task.
2. Select **Gemini 2.5 Flash** as the foundation model.

7.1.3 Provide Prompt Instructions

In the **Prompt Instructions** field, enter:

Act as a text classifier and detect the sentiment with an explanation.

Note: You may add additional instructions depending on your desired output.

7.1.4 Test Sentiment Classification

Example 1

Prompt

My least preferred fruit is mango.

Expected Output

Sentiment: Negative

Example 2

Prompt

Yesterday's papaya was delicious.

Expected Output

Sentiment: Positive

Example 3

Prompt

Despite the challenges, the team's perseverance is admirable.

Expected Output

Sentiment: Positive

7.1.5 Practice with Additional Prompts

Try the following prompts:

I am so glad to see you today.
The tech conference in Berlin was amazing.
I feel sad to see any biker fall in the motor race.

The restaurant has a good rating, but the food is below average.

After each prompt:

- Review the detected sentiment.
- Read the explanation provided by the model.
- Optionally use **Regenerate Response** to compare different outputs.

7.1.6 Learning Outcome

After completing this exercise, we should be able to:

- Perform text classification using a foundation model.
- Detect positive, negative, and mixed sentiments.
- Understand how prompt instructions influence response quality.

7.2 Exercise 2: Get Answers to Your Questions Using a Foundation Model

7.2.1 Objective

Use the **GPT-5 Nano** foundation model to answer questions using natural language.

7.2.2 Set Up the AI Environment

1. Create or rename your chat.
2. Select **GPT-5 Nano** as the foundation model.

7.2.3 Provide Prompt Instructions

In the **Prompt Instructions** field, enter:

Act as a knowledgeable resource and respond to the question.

Note: Prompt Instructions are optional, but help guide the model's responses.

7.2.4 Ask Your First Question

Prompt

What is the longest word in English?

Review the response generated by the model.

7.2.5 Ask Additional Questions

You may continue in the same chat or start a new chat.

Example Prompt

What English language statement encompasses all alphabets?

Review the generated response.

7.2.6 Practice with Your Own Questions

Ask additional questions on any topic, such as:

What is Artificial Intelligence?
Explain the difference between Machine Learning and Deep Learning.
What are Foundation Models?
How does a Transformer model work?
What are the applications of Generative AI?

After each response:

- Review the answer.
- Ask follow-up questions if needed.
- Use **Regenerate Response** to compare alternative answers.

7.2.7 Learning Outcome

After completing this exercise, I should be able to:

- Use GPT-5 Nano for question answering.
- Write effective prompts to obtain accurate responses.
- Continue multi-turn conversations while maintaining context.
- Improve responses through prompt refinement and follow-up questions.

8 Other Foundation Models

8.1 Introduction

Foundation models are large-scale AI models trained on vast amounts of data that can be adapted to perform a wide range of tasks. The following table summarizes several prominent foundation models and their distinguishing features.

8.2 Distinguishing Features of Various Foundation Models

Model	Distinguishing Features
GPT-5 / GPT-5 Mini / GPT-5 Nano (OpenAI)	The GPT-5 family (released August 2025) is OpenAI's first unified model system, combining fast and deep reasoning modes with an automatic router that selects the appropriate thinking depth for each request. All three models support multimodal inputs and a **400K-token context window** (maximum amount of text the model can process at once). **GPT-5** is the flagship model for complex reasoning, coding, and health-related tasks. **GPT-5 Mini** balances capability, speed, and cost for conversational AI and moderately complex workflows. **GPT-5 Nano** is optimized for ultra-low latency, making it ideal for classification, data extraction, and lightweight AI agents where speed and cost are critical.

GPT-5.4 / GPT-5.4 Mini / GPT-5.4 Nano (OpenAI)	Released in **March 2026**, GPT-5.4 is OpenAI's first general-purpose model with **native computer-use capabilities** (ability to control a computer using screenshots, mouse, and keyboard commands). It supports a **1 million-token context window** in the API. **GPT-5.4** targets professional knowledge work and advanced AI agent workflows. **GPT-5.4 Mini** operates approximately twice as fast as GPT-5 Mini while approaching GPT-5.4's coding performance, making it suitable for latency-sensitive applications and coding assistants. **GPT-5.4 Nano** is designed for edge devices and high-volume tasks where speed and low cost are the primary priorities.
GPT-5.5 (OpenAI)	Released in **April 2026**, GPT-5.5 is OpenAI's flagship model for long-horizon agentic workflows (tasks requiring planning, tool use, self-verification, and multi-step reasoning over extended interactions). It matches GPT-5.4's response speed while delivering higher intelligence and using fewer tokens for equivalent tasks. It excels in software development, online research, data analysis, document creation, and operating software through tools.
Mistral Small 3.1 24B (Mistral AI)	A compact enterprise-grade **24-billion-parameter** model supporting multimodal inputs (text and images), function calling, and a **128K-token context window**. It is well suited for instruction following, conversational assistants, multilingual question answering, summarization, classification, and Retrieval-Augmented Generation (RAG) across English, French, German, Italian, and Spanish.
Mistral Medium 3 (mistral-medium-2505) (Mistral AI)	A high-performance enterprise model that balances strong reasoning ability and multimodal support with lower cost than frontier models. It supports a **128K-token context window** and more than **80 programming languages**. It is particularly effective for programming, mathematical reasoning, long-document understanding, function calling, and AI agent workflows, while being optimized for single-node inference (running efficiently on a single computing system).
DeepSeek V4 Pro (DeepSeek)	Released in **April 2026** under the **MIT License**, this is the largest open-weight model released to date, containing **1.6 trillion parameters** with **49 billion active parameters per token**. It supports a **1 million-token context window** and introduces a **hybrid attention architecture** (an optimized attention mechanism that significantly reduces computational requirements). It offers three configurable reasoning modes: **Non-think**, **Think High**, and **Think Max**. The model demonstrates strong coding and scientific reasoning capabilities, supports self-hosting through open weights, and provides significantly lower API costs than comparable closed-source models.
Gemma 4 26B (MoE) / Gemma 4 31B (Dense) (Google)	Released in **April 2026**, both models are based on Google's Gemini 3 research and support multimodal inputs including text, images, videos, and audio. They provide configurable thinking modes, function calling, and a **256K-token context window**. **Gemma 4 26B** is a **Mixture-of-Experts (MoE)** model that activates only **4 billion parameters per token**, enabling high performance with lower GPU memory usage. **Gemma 4 31B** is a dense model designed to maximize overall quality and serve as a strong foundation for fine-tuning. Both are available as open-weight models and rank among the top open models on the Arena AI leaderboard.

Llama 4 Maverick 17B (128E) (Meta)	A natively multimodal **Mixture-of-Experts** model with **17 billion active parameters** and **400 billion total parameters**, utilizing **128 experts** and an **early fusion architecture** (combining multiple input modalities at an early stage for better integration). It supports a **1 million-token context window** and is designed for high-quality conversational AI, visual reasoning, code generation, and multilingual applications across **12 languages**.
Llama 4 Scout 17B (16E) (Meta)	A natively multimodal **Mixture-of-Experts** model with **17 billion active parameters** and **109 billion total parameters**. It features an industry-leading **10 million-token context window**, enabling analysis of entire codebases, large document collections, and persistent memory in a single session. It supports text and image inputs, 12 languages, and assistant-style interactions.
GPT Image 1.5 / GPT Image 2 (OpenAI)	Image generation and editing models supporting text-to-image creation, image editing, and highly accurate prompt following. **GPT Image 1.5** (December 2025) introduced improved prompt adherence, better text rendering within images, and generation speeds up to four times faster than previous versions. **GPT Image 2** (April 2026) adds a dedicated thinking mode for complex image composition, near-perfect text rendering, transparent background support, flexible aspect ratios, and image generation up to **4K resolution**.
IBM Granite 4 Small (IBM)	A **32-billion-parameter** hybrid **Mixture-of-Experts** model with **9 billion active parameters** per inference, built using a **Mamba-2/Transformer hybrid architecture** (combining Mamba sequence modeling with Transformer capabilities). It is designed for enterprise instruction following, tool calling, multi-step AI agent workflows, software development, summarization, classification, and Retrieval-Augmented Generation (RAG). It supports **12 languages** and is released under the **Apache 2.0 License**. The native context window is **128K tokens** (20K tokens when accessed through IBM watsonx).

9 IBM watsonx Generative AI Capabilities

9.1 Introduction

IBM watsonx offers advanced generative AI capabilities tailored to enterprise needs. It leverages state-of-the-art AI models to generate text, code, and data insights.

Its Natural Language Processing (NLP) capabilities enable the creation of detailed and contextually accurate:

- Reports
- Summaries
- Responses

These capabilities make watsonx valuable for:

- Customer service
- Content creation
- Data analysis

IBM watsonx is a complete compilation of AI tools that includes access to various GPT models and other Large Language Models (LLMs).

9.2 Core Components of IBM watsonx

The watsonx platform consists of three major components:

- watsonx.data
- watsonx.ai
- watsonx.governance

9.3 watsonx.data

The first step in solving any data science or generative AI use case is accessing the appropriate dataset.

watsonx.data helps organizations overcome **data silos** (situations where data is isolated across different systems or departments) by providing access to data regardless of where it resides.

9.3.1 Capabilities of watsonx.data

Powered by generative AI, watsonx.data allows users to perform tasks using natural language, including:

- Searching for the required dataset.
- Merging tables.
- Performing data preprocessing.
- Running interactive queries using natural language.

9.4 watsonx.ai

The next phase is using **watsonx.ai**, which provides a collection of Large Language Models (LLMs).

These include:

- IBM Large Language Models developed by IBM Research.
- Open-source Large Language Models.

These models are made available for both end users and enterprise customers.

9.4.1 High-Quality Model Training

IBM ensures that the data used to train its Large Language Models is:

- Well prepared.
- Properly cleaned.
- Free from offensive content.

- Designed to minimize bias.

9.4.2 AI Development Tools

watsonx.ai also provides several automated software tools that make it more convenient for:

- Data Scientists
- Data Engineers

These tools simplify the development, deployment, and management of AI models.

9.5 watsonx.governance

watsonx.governance helps organizations manage, monitor, and govern AI models throughout their lifecycle.

9.5.1 Model Management

The platform provides a unified dashboard that allows users to:

- Catalog AI models.
- Manage different model versions.
- Track the current stage of each model.

9.5.2 Performance Monitoring

watsonx.governance continuously monitors models for issues such as:

- Hallucinations (AI-generated information that appears convincing but is factually incorrect)
- Bias
- Other model performance problems

If issues are detected, the platform can:

- Generate alerts.
- Automatically mitigate certain problems.
- Identify the specific transaction or dataset responsible for the issue.

9.6 Additional Generative AI Capabilities

Beyond model management, watsonx also supports several advanced AI capabilities.

9.6.1 Code Generation

watsonx can generate code, helping developers:

- Automate repetitive coding tasks.

- Improve software development efficiency.

9.6.2 Synthetic Data Generation

The platform can generate **synthetic data** (artificially generated data that resembles real-world data while protecting sensitive information).

This helps:

- Train machine learning models.
- Build robust datasets.
- Address data privacy concerns.

9.7 Model Evaluation

Evaluating the outputs of Large Language Models is important because their outputs are often text or images, making evaluation more sophisticated than traditional machine learning models.

Different tasks require different evaluation methods.

For tasks such as:

- Retrieval-Augmented Generation (RAG)
- Summarization
- Text Generation

watsonx automatically includes the appropriate **evaluation metrics** (methods used to measure model performance).

When a user selects a task, the platform:

- Automatically chooses suitable evaluation metrics.
- Helps evaluate model performance over time.

9.8 Key Features of IBM watsonx

IBM watsonx provides:

- Enterprise-grade generative AI capabilities.
- Access to IBM and open-source Large Language Models.
- Natural language interaction with enterprise data.
- Automated data preprocessing.
- AI model governance and monitoring.
- Code generation support.
- Synthetic data generation.
- Built-in evaluation metrics for generative AI tasks.
- Continuous model performance monitoring.

9.9 Summary

IBM watsonx is a comprehensive enterprise AI platform that combines data access, Large Language Models, AI governance, code generation, synthetic data creation, and automated model evaluation within a single ecosystem. Its capabilities extend far beyond those expected from a basic AI platform, making it a powerful solution for developing, deploying, monitoring, and managing enterprise-scale generative AI applications.

10 Lesson Summary

At this point, we have learned about the core concepts of Generative AI, including deep learning and Large Language Models (LLMs). Key Learning Outcomes

- The core concepts of Generative AI, such as Large Language Models (LLMs), can perform human-like tasks.
- LLMs leverage the power of transformer networks to pre-train deep learning algorithms on vast datasets.
- These algorithms capture patterns and hierarchies (structured relationships within data) in datasets to generate accurate, human-like responses, making Generative AI highly scalable.
- The core generative AI models serve as the building blocks of Generative AI, with each model offering distinctive features.

10.1 Core Building Blocks of Generative AI

10.1.1 Variational Autoencoders (VAEs)

- Rapidly reduce the dimensionality (reduce the number of variables while preserving important information) of data samples.

10.1.2 Generative Adversarial Networks (GANs)

- Use competing neural networks to generate realistic data samples.

10.1.3 Transformer-Based Models

- Use attention mechanisms (techniques that allow the model to focus on the most relevant parts of the input while ignoring less important information) to model long-term text dependencies.

10.1.4 Diffusion Models

- Address information decay by removing noise from the latent space (the compressed mathematical representation of data within the model).

10.2 Foundation Models

- Foundation models are pre-trained on billions of parameters, enabling them to develop independent reasoning capabilities and perform a wide variety of complex tasks.
- Due to their multimodal (supporting multiple data types such as text, images, audio, and video) and multidomain (applicable across various fields and industries) capabilities, foundation models serve as the base for a wide range of Generative AI applications.

11 IBM Granite Foundation Models

11.1 Introduction

The rapid progress of Artificial Intelligence (AI) is making work smarter and more efficient. As the AI-for-business revolution gains momentum, organizations have significant opportunities to improve productivity and creativity.

To leverage this potential, IBM Research has developed a new family of foundation models, available on **watsonx.ai**, IBM's studio for Generative AI, foundation models, and machine learning.

This family of generative AI models is collectively called **Granite**, drawing an analogy to granite as a sturdy and versatile construction material.

Granite foundation models are built on a **decoder-only architecture**, which applies Generative AI to language and code generation.

11.2 IBM Granite Models

IBM recognizes that a single foundation model cannot satisfy the diverse requirements of every business use case. Therefore, the Granite model family is developed in multiple sizes.

The **Granite Base 13 Billion (granite.13b.base) V1.0** model has been trained on more than **1 trillion tokens**. It serves as the base model from which additional variants are fine-tuned for specific downstream tasks.

Two important fine-tuned variants are:

- **granite-13b-instruct**
- **granite-13b-chat**

Here, **13B** indicates that each model contains **13 billion parameters**.

Both variants are built on a **decoder-only architecture**, which forms the foundation of Large Language Models (LLMs) by predicting the next word in a sequence.

11.3 Granite Model Variants

11.3.1 granite-13b-instruct

The **granite-13b-instruct** variant is an **Instruction-Tuned Supervised Fine-Tuning (SFT)** model.

It has been further fine-tuned using a combination of:

- Flan Collection
- Approximately 15,000 samples from Dolly
- Anthropic's human preference data focusing on helpfulness and harmlessness
- Instructv3
- Internal synthetic datasets specifically designed for summarization and dialogue tasks (approximately 700,000 samples)

Note: Supervised Fine-Tuning (SFT) is a machine learning technique that adapts a pre-trained Large Language Model (LLM) to a specific downstream task using labeled data. The model is trained on input-output pairs, where the input is a prompt or instruction and the output is the desired response. Common downstream tasks include:

- Text classification
- Natural language inference
- Named entity recognition
- Question answering

11.3.2 granite-13b-chat

The **granite-13b-chat** variant is a **Contrastive Fine-Tuning (CFT)** model.

It uses an objective called **unlikelihood training**, which penalizes unlikely generations by assigning them lower probability values.

Note: Contrastive Fine-Tuning (CFT) trains an LLM using two examples:

- A positive example that the model should learn to recognize.
- A negative example that the model should learn to distinguish from the positive example.

The model learns representations that are similar for positive examples and dissimilar for negative examples.

11.4 Key Advantages of IBM Granite Models

The Granite foundation models provide several important advantages:

- The 13-billion-parameter models are more efficient than many larger models.
- They can fit into a single **V100-32GB Graphics Processing Unit (GPU)**.
- They support a wide range of business-domain tasks, including:
    - Summarization

    - Question answering
    - Classification
- Having been trained on more than **1 trillion tokens**, they are highly competitive and enterprise-ready.
- They include strong governance measures suitable for enterprise applications.
- Training data is rigorously examined using IBM's proprietary **Hateful and Profane (HAP) Detector**, which helps detect and remove hateful and profane content.
- They guarantee **data ownership**, allowing organizations to maintain control over the models they build and ensuring model portability.

11.5 IBM Granite Models: Business Use Cases

Granite models are specifically designed for enterprise applications and are available in multiple sizes to meet different business requirements.

Common use cases include:

- **Retrieval-Augmented Generation (RAG)** (combines external knowledge retrieval with an LLM to generate more accurate and context-aware responses) for searching enterprise knowledge bases and producing personalized customer responses.
- **Insight extraction** for identifying useful information from business data.
- **Classification** for tasks such as customer sentiment analysis.
- **Summarization** for converting lengthy documents, contracts, or call transcripts into concise summaries.
- **Content generation** for creating business-related text.
- **Named Entity Recognition (NER)** (identifying entities such as people, organizations, locations, dates, and other important information within text).
- Various other **Natural Language Processing (NLP)** tasks.

11.6 Key Takeaways

- IBM Granite is a family of enterprise-focused foundation models developed by IBM Research and available through **watsonx.ai**.
- Granite models use a **decoder-only architecture** for language and code generation.
- The Granite model series includes two major fine-tuned variants:
    - **granite-13b-instruct**
    - **granite-13b-chat**
- **granite-13b-instruct** uses **Supervised Fine-Tuning (SFT)**.
- **granite-13b-chat** uses **Contrastive Fine-Tuning (CFT)** with **unlikelihood training**.
- Granite models are trained on more than **1 trillion tokens**, making them suitable for enterprise-scale applications.
- IBM emphasizes trustworthy AI by incorporating rigorous content moderation using its proprietary **HAP detector**.
- Granite foundation models guarantee **data ownership** and support model portability.
- Common enterprise applications include Retrieval-Augmented Generation (RAG), insight extraction, classification, summarization, content generation, Named Entity Recognition (NER), and other Natural Language Processing (NLP) tasks.

Module 2: Platforms for Generative AI

12 Pre-Trained Models: Text-to-Text Generation

12.1 Introduction to Text-to-Text Generation Models

Text-to-text generation models are a type of machine learning model used to generate text from a given input.

These models are trained on a large corpus (collection) of text to learn:

- Language patterns
- Grammar
- Contextual information (information that helps understand the meaning based on surrounding text)

Using this learned knowledge, the models generate new text in various formats, including:

- Summaries
- Source code
- Translations
- Creative content
- Scripts
- Musical pieces
- Emails
- Letters
- And many other forms of text

12.2 Types of Text-to-Text Generation Models

There are two main types of text-to-text generation models:

- Statistical Models
- Neural Network Models

12.2.1 Statistical Models

Statistical models use statistical techniques to generate text.

One of the most common statistical models is the **Markov Chain**.

A Markov Chain generates text by:

- Starting with a seed state (initial input).
- Predicting the next state based on the previous state.

For example, it can predict and generate the next character based on previously observed language patterns.

12.2.1.1 Applications of Markov Chains

Markov Chains are commonly used for:

- Speech recognition
- Journalism
- Text prediction

12.2.2 Neural Network Models

Neural network models use artificial neural networks to generate text.

These models:

- Learn complex relationships within data.
- Are trained on large text corpora.
- Generate text similar to the data they were trained on.

12.3 Text Generation Architectures

Text-to-text generation models generally use one of the following architectures:

- Sequence-to-Sequence Models
- Transformer Models

12.3.1 Sequence-to-Sequence Models

Sequence-to-sequence (Seq2Seq) models work in two stages:

- The encoder converts the input text into a sequence of numbers.
- The decoder converts this numerical representation into a new sequence representing the generated text.

12.3.1.1 Common Applications

- Text summarization
- Speech recognition
- Machine translation

12.3.2 Transformer Models

Transformer models directly map the input text to the generated text.

Compared to sequence-to-sequence models, transformers generate more fluent and natural-sounding text.

A key feature of transformer models is the **Attention Mechanism** (a technique that enables the model to focus on the most relevant words or tokens while generating text).

Attention helps provide context for a specific word or token, including tokens that may represent other types of data, such as sections of an image.

12.4 Popular Text-to-Text Generation Models

Some of the most widely used text-to-text generation models include:

- GPT
- T5
- BART

12.4.1 GPT

GPT (Generative Pre-trained Transformer), developed by OpenAI, is a large language model trained on an extensive corpus of text and code.

Its capabilities include:

- Text generation
- Language translation
- Creative content generation
- Answering user questions
- Code generation

12.4.2 T5

T5 (Text-to-Text Transfer Transformer), developed by Google AI, is also trained on a large dataset of text and code.

It can perform a wide variety of Natural Language Processing (NLP) tasks, including:

- Summarization
- Translation
- Question answering

12.4.2.1 Why T5 Is Popular

Traditional NLP approaches often require separate models for different tasks, such as:

- Language translation
- Language auto-completion

This makes the overall approach inefficient.

T5 addresses this limitation by treating every NLP problem as a text-to-text task.

It converts all input and output data into text, enabling a single model to learn from diverse examples and perform many different NLP tasks.

T5:

- Uses a standard encoder-decoder architecture.
- Performs tasks such as text transformation and classification.
- Leverages **transfer learning** (reusing knowledge learned from one task to improve performance on another related task).

A model trained for one task can be fine-tuned to perform another downstream task (a task performed after pretraining within the same domain).

12.4.3 BART

BART (Bidirectional and Auto-Regressive Transformer), developed by Facebook AI, is a deep neural network based on a sequence-to-sequence architecture.

It combines:

- A bidirectional encoder similar to **BERT**.
- A left-to-right decoder similar to **GPT**.

BERT (Bidirectional Encoder Representations from Transformers) is a family of language models developed by Google that uses pretraining and fine-tuning to accomplish various NLP tasks.

12.4.3.1 Key Features of BART

- Processes text in both forward and backward directions.
- Combines the strengths of BERT and GPT.
- Generates contextually accurate text by using previously generated tokens to predict the next ones.

12.4.3.2 Applications of BART

BART can perform:

- Sentiment analysis
- Question answering
- Language translation
- Human-like text generation
- Computer vision tasks

Its versatility makes it valuable across multiple industries.

12.5 Applications of Text-to-Text Generation Models

Text-to-text generation models are powerful tools for automating a wide variety of tasks.

12.5.1 Text Summarization

These models can process lengthy input text and generate concise summaries without changing the original meaning.

12.5.2 Conversational Intelligence

They support conversational intelligence by providing personalized assistance through text-based query responses.

12.5.3 Content Creation

They can generate digital text for tasks such as:

- Product descriptions
- Emails
- Resumes
- Marketing content
- Social media posts

12.6 Benefits of Text-to-Text Generation Models

Text-to-text generation models offer several important benefits.

12.6.1 Increased Productivity

They automate repetitive writing tasks, such as:

- Creating marketing copy
- Generating social media posts
- Producing business content

12.6.2 Improved Accuracy

They improve the accuracy of tasks that are prone to human error.

For example, they can translate documents while preserving the intended context and meaning.

12.7 Key Takeaways

- Text-to-text generation models generate text from a given input after learning language patterns from large text datasets.
- There are two primary types of text-to-text generation models: **Statistical Models** and **Neural Network Models**.
- These models typically use either **Sequence-to-Sequence** or **Transformer** architectures.
- **GPT** is a powerful large language model capable of text generation, translation, coding, and question answering.

- **T5** treats every NLP problem as a text-to-text task and supports multiple tasks using a single unified model.
- **BART** combines the strengths of BERT and GPT to perform a wide range of NLP and computer vision tasks.
- Text-to-text generation models are widely used for summarization, conversational AI, content creation, translation, and many other real-world applications.
- These models improve productivity through automation and increase accuracy in text-based tasks.

13 Developing AI Applications with Generative AI Foundation Models

13.1 Introduction

Generative AI foundation models provide powerful natural language understanding and generation capabilities, enabling the development of AI applications for tasks such as text generation, vocabulary enhancement, language learning, translation, summarization, and other Natural Language Processing (NLP) applications.

Choosing an appropriate foundation model based on the use case is an important step in designing efficient and effective AI-powered solutions.

13.2 Learning Objectives

After completing these exercises, we are able to:

- Explore vocabulary generation using **GPT-5 Nano** to develop AI-powered dictionary or vocabulary-learning applications.
- Utilize **GPT-5 Nano** for interactive language learning and educational applications.
- Explore the translation capabilities of foundation models to build AI-powered translation services.

13.3 Exercise 1: Vocabulary Generation and Language Learning

This exercise demonstrates how **GPT-5 Nano** can be used to improve vocabulary and support interactive language learning.

13.3.1 Foundation Model

- **GPT-5 Nano**

13.3.2 Prompt Template

System Instruction

Act as a language expert. Start a conversation in the specified language, explain each concept clearly, and continue the learning process by asking follow-up questions.

Example User Prompt

English

13.3.3 Practical Applications

Using appropriate prompts, the model can:

- Explain the meaning of words.
- Generate synonyms and antonyms.
- Describe appropriate and inappropriate word usage.
- Create fill-in-the-blank questions and multiple-choice quizzes.
- Conduct interactive conversations to reinforce vocabulary learning.

These capabilities can be utilized to build:

- Vocabulary trainers
- AI dictionaries
- Educational assistants
- Personalized language-learning applications

13.4 Exercise 2: Learning a Foreign Language

This exercise demonstrates how **GPT-5 Nano** can be used as an interactive language tutor for learning a foreign language.

13.4.1 Foundation Model

- **GPT-5 Nano**

13.4.2 Prompt Template

System Instruction

Act as a language expert. Communicate in the specified language, explain vocabulary and grammar clearly, and encourage continuous learning through follow-up questions.

Example User Prompt

German

13.4.3 Practical Applications

The model can assist with:

- Learning vocabulary in a foreign language.
- Explaining word meanings and usage.
- Providing synonyms and antonyms.
- Teaching idioms and common expressions.
- Creating practice exercises and quizzes.
- Supporting conversational language practice.

These capabilities enable the development of:

- AI-powered language tutors
- Educational platforms
- Personalized language-learning applications

13.5 Exercise 3: Language Translation

This exercise explores how **GPT-5 Nano** performs multilingual text translation and supports AI-powered translation applications.

13.5.1 Foundation Model

- **GPT-5 Nano**

13.5.2 Prompt Template

System Instruction

Act as a professional translation expert and accurately translate text from the source language into the target language while preserving the original meaning and context.

Example User Prompts

English → German

or

English → French

13.5.3 Practical Applications

The model can:

- Translate text between multiple languages.
- Preserve meaning, tone, and context during translation.
- Translate both short and long-form content.
- Support multilingual communication.
- Assist in building AI-powered translation services and multilingual applications.

13.6 Summary

By completing these exercises, we explored how **GPT-5 Nano** can be used to develop AI applications for vocabulary enhancement, language learning, and multilingual translation.

Through carefully designed prompts, foundation models can:

- Generate educational content.
- Support conversational learning.
- Explain language concepts.

- Perform accurate text translation.

These capabilities extend to numerous Natural Language Processing (NLP) applications, including:

- AI tutors
- Vocabulary assistants
- Educational platforms
- Multilingual chatbots
- Content generation systems
- Translation services

In real-world applications, these capabilities can be integrated programmatically using the **OpenAI API**, enabling developers to build intelligent language-based AI solutions.

14 Pre-trained Models: Text-to-Image Generation

14.1 Introduction to Text-to-Image Generation Models

Text-to-image generation models are a type of machine learning model used to generate images from text descriptions.

They use Generative AI to make meaning out of words and transform them into unique images.

These models are trained on a large dataset of text and images and can generate various types of images, including:

- Realistic images
- Abstract images
- Images that do not exist in the real world

There are two main types of text-to-image generation models:

- Generative Adversarial Networks (GANs)
- Diffusion Models

14.2 Generative Adversarial Networks (GANs)

GANs are a type of machine learning model that can generate realistic images.

They work by training two models against each other:

- **Generator Model**
    - Generates images.
- **Discriminator Model**
    - Distinguishes between real and fake images.

As training progresses, the generator model is gradually improved until it can generate images that are nearly indistinguishable from real images.

14.3 Diffusion Models

Diffusion models are another type of machine learning model used to generate images from text descriptions.

They work by:

- Starting with a random image.
- Gradually adding detail to the image until it matches the provided text description.

Compared to GANs, diffusion models are typically:

- More efficient.
- Better at generating creative and abstract images.

Among diffusion models, two of the most popular text-to-image generation models are:

- DALL·E
- Imagen

14.4 DALL·E

DALL·E is a text-to-image generation model developed by OpenAI.

It is trained on a massive dataset of text and images and can generate realistic images from a wide variety of text descriptions.

DALL·E also incorporates an evaluation mechanism to determine whether the final generated image accurately represents the given prompt.

To achieve this, DALL·E combines:

- Natural Language Processing (NLP)
- Machine Learning
- Computer Vision

For example, it can take a simple description such as:

"A panda juggling a ball."

and generate a photorealistic image that has never existed before.

Another important capability of DALL·E is **inpainting** (editing specific parts of an existing image based on a text description while blending the edits seamlessly with the original image).

DALL·E is unique because deep learning enables it to understand:

- Individual objects (such as pandas and balls).
- The relationships between those objects.

Interestingly, the latest version of DALL·E is a large model, but it is:

- Smaller than GPT-3.
- Smaller than its predecessor.

Despite its smaller size, the newer version generates images with **four times better resolution** than the previous version.

14.4.1 Benefits of DALL·E

DALL·E offers several important benefits:

- It demonstrates whether the AI system truly understands the user's instructions rather than simply repeating patterns it has learned.
- It helps researchers develop safer and more useful AI by improving our understanding of how AI interprets the world.

14.5 Imagen

Imagen is a text-to-image generation model developed by Google AI.

Like DALL·E, Imagen is trained on a massive dataset of text and images.

It is designed to generate realistic images from a wide variety of text descriptions.

Imagen leverages:

- Large transformer models.
- Deep language understanding.

These capabilities enable it to generate **high-fidelity images** (images with exceptional detail, quality, and realism).

An important discovery behind Imagen is that large language models, such as **T5** (a transformer model pre-trained on text), are highly effective at encoding text for image generation.

Based on this finding, Imagen increases the size of its language model rather than increasing the size of the image diffusion model, resulting in improved image fidelity (the accuracy and realism of the generated image).

One of Imagen's key advantages is its ability to produce an unprecedented degree of photorealism.

For example, the model can interpret a prompt such as:

"Two white umbrellas rising into the sky. Clouds are moving. It is raining."

and generate an image depicting exactly that scene.

The generated image may be:

- Photorealistic.
- An artistic interpretation of the prompt.

14.6 Key Takeaways

- Text-to-image generation models convert text descriptions into images using Generative AI.
- These models are primarily classified into two categories:
    - Generative Adversarial Networks (GANs)
    - Diffusion Models
- GANs generate realistic images through competition between a generator and a discriminator.
- Diffusion models gradually transform random noise into images that match a text description.
- DALL·E combines Natural Language Processing, Machine Learning, and Computer Vision to generate and edit realistic images.
- DALL·E supports **inpainting**, allowing seamless image editing using text prompts.
- Imagen uses large transformer models and deep language understanding to generate highly realistic, high-fidelity images.
- Both DALL·E and Imagen represent significant advances in text-to-image generation and continue to expand the capabilities of Generative AI.

15 Pre-trained Models – Text-to-Code Generation

15.1 Learning Objectives

After studying this topic, we will be able to:

- Explain how text-to-code generation models work in Generative AI.
- Identify the different text-to-code generation models.
- Describe the uses and benefits of text-to-code generation models and tools.

15.2 Introduction to Text-to-Code Generation

Text-to-code generation models are a type of machine learning model used to generate code from natural language descriptions.

These models use Generative AI to write code through **neural code generation**, which is a process that uses artificial neural networks (loosely based on the neural networks in the human brain).

These neural networks are trained on a substantial dataset of code examples and are then fine-tuned to generate:

- Code snippets
- Functions
- Complete applications

The generated code is similar in structure and function to the examples on which the model has been trained.

15.3 Types of Text-to-Code Generation Models

There are two main text-to-code generation models:

- Sequence-to-Sequence Models
- Transformer Models

15.3.1 Sequence-to-Sequence Models

Sequence-to-sequence models are machine learning models that can translate information from one domain to another.

In text-to-code generation, these models translate natural language descriptions into code sequences.

15.3.2 Transformer Models

Transformer models are machine learning models that can learn long-range dependencies (relationships between words that may be far apart in a sentence) between words.

For text-to-code generation, these models learn the relationships between the words in a natural language description and the corresponding code.

15.4 Popular Text-to-Code Generation Models

Several popular models belong to the sequence-to-sequence and transformer model categories.

15.4.1 CodeT5

CodeT5 is a sequence-to-sequence model developed by Google AI.

It is the first pre-trained programming language model that is:

- Code-aware
- Encoder-decoder based

Like most models, CodeT5 is trained on a large dataset of text and code.

It enables a wide range of code intelligence applications, including:

- Code understanding
- Code generation
- Text summarization
- Question answering
- Language translation

15.4.2 CodeToSequence

CodeToSequence is a sequence-to-sequence model developed by OpenAI.

It leverages the syntactic structure (the grammatical structure of programming languages) of programming languages to encode source code more effectively.

After being trained on a substantial text and code dataset, CodeToSequence can generate code for tasks such as:

- Code summarization
- Documentation
- Code retrieval

For example, CodeToSequence can caption code snippets by assigning a natural language description to the task performed by the snippet.

15.4.3 PanGuCoder

PanGuCoder is a transformer model developed by Microsoft Research.

It is a pre-trained decoder-only language model that generates code from natural language descriptions.

PanGuCoder is built on the **PanGu-Alpha architecture**, a large-scale neural network designed for Natural Language Processing (NLP).

It can generate code for tasks such as:

- Function definition
- Class definition
- Program synthesis (automatically generating complete programs)

For example, PanGuCoder can synthesize executable code based on a natural language description of a problem.

15.5 Other Text-to-Code Foundation Models and Tools

Several general-purpose foundation models can also be used for text-to-code generation.

15.5.1 OpenAI GPT

GPT excels in human-like text generation and demonstrates impressive capabilities in code creation.

15.5.2 CodeLlama

CodeLlama, developed by Meta, can:

- Generate code
- Explain code in natural language (specifically English)

15.5.3 GitHub Copilot

GitHub Copilot is powered by OpenAI Codex.

It can generate code for various programming languages and frameworks.

15.5.4 IBM Watson Code Assistant

IBM Watson Code Assistant is built on IBM watsonx.ai.

It enables developers to write code accurately and efficiently through features such as:

- Real-time recommendations
- Autocomplete
- Code restructuring assistance

15.6 Uses and Benefits of Text-to-Code Generation Models

Text-to-code generation models and tools provide several practical benefits for software development.

15.6.1 Auto-Completion

Text-to-code generation plays a crucial role in autocompletion by suggesting and completing code snippets or statements as developers type.

For example:

- As a developer writes a code comment in Python or another programming language, the model suggests relevant keywords, variable names, or function names based on the context.
- This improves code documentation and readability.

15.6.2 Automatic Code Generation

Text-to-code generation models can automatically generate code from natural language descriptions.

Example

Task:

Generate a Python function that calculates the factorial of a given integer.

The model can generate the corresponding Python code automatically.

15.6.3 Debugging Assistance

The stable underlying architecture of text-to-code generation tools helps reduce the time spent on debugging (identifying and fixing errors in software programs).

Example

Suppose a developer working on a chatbot notices that:

"The chatbot gives irrelevant responses to user queries. I need to fine-tune the model."

The text-to-code generation model generates code snippets for fine-tuning the chatbot using user feedback data.

This helps the developer quickly improve the chatbot's performance.

15.6.4 Code Translation

Text-to-code generation models can automatically translate code between different programming languages, improving cross-platform compatibility and simplifying software migration.

15.6.5 Code Refactoring and Application Modernization

These models assist with:

- Code refactoring (improving existing code without changing its functionality)
- Application modernization

They automatically identify outdated or inefficient code segments and suggest optimized replacements.

15.6.6 Library and Framework Recommendations

Text-to-code generation models recommend suitable libraries and frameworks based on project requirements, helping developers make informed technology choices.

15.6.7 Test Data Generation

These models can automatically generate code that populates databases or data structures with diverse and realistic test cases, reducing the time required for software testing.

15.6.8 Code Documentation

Text-to-code generation models can automatically generate:

- Code comments
- Function descriptions
- Documentation

This makes codebases easier to understand and maintain.

15.7 Key Takeaways

- Text-to-code generation models generate code from natural language descriptions using neural code generation.
- There are two primary categories of text-to-code generation models:
  - Sequence-to-Sequence Models
  - Transformer Models
- Popular text-to-code generation models include:
  - CodeT5
  - CodeToSequence
  - PanGuCoder
- General-purpose foundation models and tools include:
  - OpenAI GPT
  - CodeLlama
  - GitHub Copilot
  - IBM Watson Code Assistant
- Text-to-code generation models provide numerous benefits, including:
  - Auto-completion
  - Automatic code generation
  - Debugging assistance
  - Code translation
  - Code refactoring
  - Application modernization
  - Library and framework recommendations
  - Test data generation
  - Automatic code documentation

16 Developing AI Applications for Code Generation

16.1 Learning Objectives

After completing this topic, we will be able to:

- Generate code in the desired programming language using **GPT-5 Nano**.
- Explore how foundation models can assist in developing AI-powered programming applications.
- Understand how prompt engineering influences code generation quality and accuracy.

16.2 Introduction

Generative AI foundation models can generate code for a wide range of programming tasks, enabling developers to build AI-powered programming assistants, automate software development tasks, and accelerate the coding process.

Selecting an appropriate foundation model for code generation helps create efficient, accurate, and context-aware AI applications.

16.3 Generating Code Using a Foundation Model

This topic demonstrates how **GPT-5 Nano** can generate source code from natural language instructions, making it useful for software development, programming education, and AI-assisted coding applications.

16.3.1 Foundation Model

- **GPT-5 Nano**

16.3.2 Prompt Template

System Instruction

Act as a coding expert and generate accurate, well-structured code in the specified programming language based on the given requirements.

Example User Prompt

Python

Task Prompt

Write complete Python code to create a Fibonacci series up to 10.

16.4 Practical Applications

Using appropriate prompts, the model can:

- Generate code in multiple programming languages.
- Produce complete implementations for programming tasks and algorithms.
- Explain coding concepts and programming logic.
- Assist in learning programming through code examples and step-by-step guidance.
- Accelerate software development by generating boilerplate code and solving common programming problems.

16.5 Code Validation

Although foundation models can generate functional code, the generated output should always be tested and validated before deployment.

Running the generated code in a compiler or development environment helps verify its correctness, identify potential errors, and ensure it satisfies the intended requirements.

In addition, AI-generated code should be reviewed for:

- Factual accuracy
- Efficiency
- Security
- Ethical use

While foundation models are highly capable coding assistants, they may have limitations when generating large, complex, or highly specialized software projects. Therefore, human review remains an essential part of the development process.

16.6 Summary

In this topic, I explored how **GPT-5 Nano** can generate source code from natural language prompts to support software development and programming education.

Through effective prompt engineering, foundation models can assist in:

- Writing algorithms
- Generating code in multiple programming languages
- Explaining programming concepts
- Accelerating application development

These capabilities enable the development of:

- AI-powered coding assistants
- Educational platforms
- Automated code generation tools
- Intelligent software development solutions

In real-world applications, these capabilities can be integrated programmatically using the **OpenAI API**, allowing developers to build scalable AI-powered programming applications.

17 Generative AI: Foundation Models and Platforms

17.1 Learning Objectives

After completing this reading, we will be able to:

- Identify the key contributors in the field of Generative AI.
- Describe the role of the key contributors in the field of Generative AI.
- List some of the major products and services of the key contributors in the field of Generative AI.
- Recognize the role of smaller contributors in the field of Generative AI.

17.2 Introduction

Generative AI is a branch of artificial intelligence that has the ability to create new content based on existing data such as images, text, audio, or video.

Google, IBM, Microsoft, and OpenAI are some of the key players in the generative AI landscape. They provide diverse approaches and solutions for applications and use cases such as content creation, data augmentation, privacy, personalization, and entertainment.

17.3 Google

Google has been using Generative AI to create new experiences for its users.

17.3.1 Key Products and Services

- **Gemini by Google** is an AI chatbot that can generate responses to text prompts. Like OpenAI's ChatGPT, Gemini is also a large conversational language model. However, it can access the internet to leverage Google Search and generate responses accordingly.
- **Google's Generative AI App Builder** helps developers build and deploy chatbots and search applications in minutes.
- **Duet AI for Google Workspace** makes work more effective by providing AI-powered assistance.
- **Duet AI for Google Cloud** serves as an always-on AI collaborator, offering generative AI-powered assistance to cloud users of all types.

17.4 IBM

IBM is a leading enterprise in developing and applying Generative AI models across diverse domains, including foundation models, enterprise applications, and mainframe applications.

17.4.1 Key Products and Services

- **watsonx** is a generative AI-assisted platform that accelerates the modernization of mainframe applications.
- **IBM Watson Code Assistant** uses IBM Watson AI-generated recommendations to simplify code writing. It is powered by the IBM watsonx foundation models and is designed to make IT automation accessible to everyone, not just subject matter experts.
- IBM plans to expand the **Watson Code Assistant** portfolio to support multiple programming languages and address the challenges developers face during modernization efforts.

17.5 Microsoft

Microsoft has launched several products and services that enable developers, researchers, and organizations to use Generative AI for various purposes.

17.5.1 Key Products and Services

- Microsoft has partnered with **OpenAI** to optimize the use of **Microsoft Azure**, a cloud computing platform that provides services and tools for building, deploying, and managing applications through global data centers.
- **Image Creator by Microsoft Bing** is a generative AI tool that creates images from text descriptions using AI.
- **Microsoft AI Skills Initiative** provides training and resources that empower the workforce to use Generative AI effectively.

- **Generative AI Skills Grant Challenge** supports nonprofits, social enterprises, research organizations, and academic institutions in exploring, developing, and implementing Generative AI solutions.

17.6 OpenAI

OpenAI is a nonprofit organization focused on AI research and development. Its generative AI models can produce language, code, and images.

17.6.1 Key Products and Services

- **Generative Pre-trained Transformer (GPT)** is a language generation model capable of producing human-like text. It is widely used to create short stories, social media content, and chatbots.
- **ChatGPT** is a chatbot powered by a Large Language Model (LLM). It interacts conversationally with users by answering follow-up questions, admitting mistakes, challenging incorrect assumptions, and rejecting inappropriate requests.
- **Jasper** is an AI-powered platform that helps automate tasks and reduce human labor costs. It enables organizations to efficiently integrate OpenAI technology into their operations.
- **Codex** is a model that understands and executes simple natural language commands, making it possible to build natural language interfaces for existing applications.
- **DALL·E** creates realistic images and artwork from natural language descriptions.

17.7 Other Noteworthy Contributors

Several other organizations and researchers have also made significant contributions to the advancement of Generative AI.

17.7.1 Meta Platforms, Inc.

Meta Platforms, Inc. (formerly Facebook) has made significant advances in AI algorithms.

- It uses **Neuro-Symbolic AI** (an approach that combines neural networks with symbolic reasoning) to solve real-world problems in healthcare, finance, and other domains.
- **Large Language Model Meta AI (LLaMA)** is a family of large language models designed to power applications similar to ChatGPT and Bing Chat.

17.7.2 Ian Goodfellow

Ian Goodfellow, a former Google Brain research scientist and Director of Machine Learning at Apple, created the first implementation of **Generative Adversarial Networks (GANs)**.

17.7.3 Diederik Kingma and Max Welling

Diederik Kingma, a scientist at Google, and Max Welling, a professor at the University of Amsterdam, introduced **Variational Autoencoders (VAEs)**.

VAEs are generative AI algorithms that:

- Generate new content.
- Identify anomalies.
- Remove noise using deep learning.

Both **GANs** and **VAEs** can be applied to use cases such as customer service and technical support chatbots.

17.7.4 Adobe

Adobe developed **Firefly**, a generative AI-powered content creation tool that enables users to imagine, experiment, and create an unlimited variety of images using simple text prompts in more than one hundred languages.

17.7.5 Carnegie Mellon University

Researchers at Carnegie Mellon University developed **PolyCoder**, an open-source code generation model.

PolyCoder:

- Generates code in multiple programming languages.
- Writes code from natural language prompts.
- Helps reduce development costs.
- Enables developers to focus more on creative tasks.

17.7.6 Stability AI

Stability AI created **Stable Diffusion**, a deep learning text-to-image model based on diffusion techniques.

It generates highly detailed images from text descriptions.

17.7.7 Midjourney Inc.

Midjourney Inc., an independent research lab based in San Francisco, developed **Midjourney**, one of the most popular AI software programs for creating AI-generated artwork from text prompts.

17.7.8 Hugging Face

Hugging Face is an open-source data science and machine learning platform that provides infrastructure for hosting, training, and deploying AI models.

It was founded by French entrepreneurs:

- Clement Delangue
- Julien Chaumond

- Thomas Wolf

17.7.9 Craiyon

Craiyon is a free AI image generator that produces images based on text input.

Users simply enter keywords, and Craiyon generates corresponding images.

With such a powerhouse of contributors—both large enterprises and smaller research-focused organizations—Generative AI is positioned for explosive growth, bringing unprecedented speed, creativity, and effectiveness to education, work, and society.

17.8 Summary

In this reading, we learned that **Google, IBM, Microsoft, and OpenAI** are the key players in Generative AI and explored their major contributions.

- **Google** contributes technologies such as Gemini, Generative AI App Builder, and Duet AI.
- **IBM** has developed watsonx, IBM Watson Code Assistant, and other enterprise AI solutions.
- **Microsoft** offers Image Creator by Bing, the AI Skills Initiative, and the Generative AI Skills Grant Challenge.
- **OpenAI** has created GPT, ChatGPT, Jasper, Codex, and DALL·E.

I also learned about the contributions of several other important organizations and researchers, including:

- Meta (LLaMA)
- Ian Goodfellow (GANs)
- Diederik Kingma and Max Welling (VAEs)
- Adobe Firefly
- PolyCoder
- Stable Diffusion
- Midjourney
- Hugging Face
- Craiyon

18 IBM watsonx.ai

18.1 Learning Objectives

After studying this topic, we will be able to:

- Explain the capabilities and features of IBM watsonx.ai.
- Identify the common tools available in IBM watsonx.ai.
- Understand how watsonx.ai supports the AI lifecycle for building enterprise AI applications.

18.2 Introduction to IBM watsonx

Generating a poem, artwork, or a song using AI is exciting, but when AI is applied to businesses, it must meet much higher standards.

AI solutions for businesses need to be:

- Trusted
- Secure
- Scalable
- Adaptable
- Built according to business requirements

IBM watsonx is an integrated AI and data platform designed to help AI builders develop enterprise-grade AI solutions.

18.3 Products of the IBM watsonx Platform

The IBM watsonx platform consists of three products:

- **watsonx.ai** – A studio for foundation models, Generative AI, and Machine Learning.
- **watsonx.data** – A data store for managing enterprise data.
- **watsonx.governance** – A toolkit for AI monitoring and governance.

This topic focuses on **watsonx.ai**.

18.4 What is IBM watsonx.ai?

IBM watsonx.ai is a studio of integrated tools powered by foundation models for working with Generative AI and building Machine Learning models.

Using watsonx.ai, users can:

- Train foundation models.
- Tune foundation models.
- Deploy AI models.
- Manage foundation models.

These capabilities help build AI applications in a fraction of the time and with a fraction of the data.

18.5 Goals of IBM watsonx.ai

With watsonx.ai, users can achieve several important goals, including:

- Building Machine Learning models.
- Experimenting with foundation models.
- Managing the complete AI lifecycle.

Based on these goals, users can choose different tasks available on the platform.

18.6 AI Lifecycle in watsonx.ai

The tasks and tools provided by watsonx.ai align with the AI lifecycle.

The typical workflow includes:

1. Prepare data.
2. Build an experiment.
3. Train models and AI solutions.
4. Deploy models.
5. Build AI applications.
6. Manage deployed models.
7. Automate repetitive processes.

18.7 Foundation Models in watsonx.ai

watsonx.ai provides access to:

- IBM's selected open-source models from Hugging Face.
- IBM-trained foundation models of different sizes and architectures.

As an AI value creator, users can also bring their own models and datasets into watsonx.ai.

18.8 Prompt Lab

Prompt Lab enables AI builders to experiment with foundation models and create prompts tailored to their requirements.

It supports a variety of Natural Language Processing (NLP) tasks, including:

- Question answering
- Content generation
- Summarization
- Text classification
- Information extraction

Prompt Lab provides easy-to-use tools for building and refining high-performing prompts to achieve the desired results.

18.9 Tuning Studio

Organizations often need to customize foundation models using their own business data and specific use cases.

watsonx.ai provides the **Tuning Studio** for this purpose.

Using Tuning Studio, users can:

- Import custom datasets.
- Tune models using as few as **100 examples**.
- Apply state-of-the-art tuning methods with only a few clicks.

Future versions of watsonx.ai will include additional tuning methods and examples for:

- Prompt tuning
- Fine-tuning foundation models

These techniques improve model performance and accuracy.

18.10 Pipeline Tool

Deploying an AI model into a production application is a multi-step process.

The **Pipeline Tool** helps manage and automate the model lifecycle.

It automates tasks such as:

- Loading data
- Training models
- Deploying models
- Evaluating models

Benefits of the Pipeline Tool include:

- Reduced time to move models into production.
- Improved model accuracy.
- Increased model reliability.
- Streamlined AI workflows.

18.11 Collaborative AI Development

watsonx.ai provides a collaborative environment that streamlines workflows throughout the AI lifecycle.

Foundation models allow organizations to build powerful AI applications in a fraction of the time and with significantly less data than traditional AI approaches.

The overall workflow includes:

- Building and refining prompts using Prompt Lab.
- Customizing models using Tuning Studio.
- Deploying enterprise-grade AI models.
- Building AI-powered applications.

This enables organizations to multiply the power of AI across the enterprise while keeping the development process practical, efficient, and easy to use.

18.12 Security and Privacy

IBM watsonx.ai is designed with enterprise-grade security and privacy.

Key security features include:

- Data is accessible only to the account owner.
- Data is stored in encrypted format.
- Models created by users remain private to their accounts.
- IBM does not have access to user data or models.
- User data and models are never used by IBM or any other person or organization without explicit permission.

18.13 Key Takeaways

- **IBM watsonx** is an integrated AI and data platform that helps businesses build AI responsibly and transparently.
- **watsonx.ai** is one of the three products in the watsonx platform and provides integrated tools for training, tuning, deploying, and managing Generative AI and Machine Learning models.
- The platform supports the complete AI lifecycle, from data preparation to model deployment and management.
- Users can access IBM foundation models, selected open-source models from Hugging Face, and also bring their own models and datasets.
- **Prompt Lab** is used for experimenting with foundation models and creating prompts for NLP tasks.
- **Tuning Studio** enables organizations to customize foundation models using their own data for improved performance.
- The **Pipeline Tool** automates model training, deployment, evaluation, and lifecycle management.
- IBM watsonx.ai provides enterprise-grade security, ensuring that user data and models remain private, encrypted, and under the user's control.

19 IBM watsonx.data and IBM watsonx.governance

19.1 Learning Objectives

After studying this topic, we will be able to:

- Describe the features, capabilities, and applications of **IBM watsonx.data**.
- Describe the features, capabilities, and applications of **IBM watsonx.governance**.

19.2 Introduction

IBM **watsonx** is an AI and data platform designed for businesses to enhance the impact of Artificial Intelligence (AI). It consists of three core components:

- **watsonx.ai** – Combines generative AI capabilities powered by foundation models and traditional machine learning into a powerful studio that supports the entire AI lifecycle.
- **watsonx.data** – A fit-for-purpose datastore designed for efficient data storage and management.
- **watsonx.governance** – A powerful toolkit for AI governance, monitoring, and compliance.

As a cloud-based platform, IBM watsonx offers the following advantages:

- Accessible from anywhere.
- Suitable for businesses of all sizes.
- Provides scalability to support AI integration across an organization.
- Enables businesses to adapt to evolving needs.
- Advances trustworthy AI by:
    - Enhancing data accessibility.
    - Implementing governance.
    - Reducing costs.
    - Accelerating the deployment of high-quality AI models.

This topic focuses on **watsonx.data** and **watsonx.governance**.

19.3 IBM watsonx.data

19.3.1 Overview

IBM **watsonx.data** is an extensive, carefully curated data repository equipped with a state-of-the-art data management system.

It is built on an **open lakehouse architecture** (an architecture that combines the flexibility of data lakes with the performance and management capabilities of data warehouses).

19.3.2 Features and Capabilities

Watsonx.data provides the following features:

- Can be used to train and fine-tune AI models.
- Enables effortless connection to organizational data.
- Improves data reliability through:
    - Built-in governance.
    - Security.
    - Automation features.
- Provides fast and efficient data processing through multiple query engines.
- Uses multiple storage tiers and query engines to optimize workloads and reduce data warehouse costs.
- Leverages the foundation models of **watsonx.ai** to simplify and accelerate data interaction.

Through the **watsonx.ai conversational user interface**, users can quickly:

- Retrieve data.

- Explore data.
- Augment data and metadata (additional information that describes data).

19.4 IBM watsonx.governance

19.4.1 Overview

AI is often considered a **black box** (a system whose internal decision-making process is difficult to understand), raising more questions than it resolves.

IBM **watsonx.governance** enables organizations to:

- Govern AI.
- Scale AI responsibly.
- Improve transparency into AI systems.

It is a powerful toolkit that allows organizations to direct, manage, and monitor AI activities.

19.4.2 Features and Capabilities

Watsonx.governance helps organizations design AI based on:

- Responsibility.
- Transparency.
- Trust.

It also enables organizations to:

- Trace datasets, models, and pipelines.
- Consistently explain AI-generated decisions.
- Monitor AI models for:
    - Fairness.
    - Bias.
    - Drift (changes in model performance or data patterns over time).
- Take corrective actions when necessary.
- Translate regulatory requirements into enforceable policies for effective compliance management.
- Gain visibility into AI models and processes through customizable:
    - Dashboards.
    - Reports.
    - Charts.

19.5 Applications of watsonx.governance

19.5.1 Credit Risk

Watsonx.governance enables organizations to:

- Optimize profitability.
- Implement fair, unbiased, and transparent approval processes.

19.5.2 Health Monitoring and Diagnosis

In the healthcare sector, watsonx.governance helps:

- Ensure precision in AI-assisted healthcare.
- Prevent inaccurate and potentially harmful recommendations for diagnosis and treatment.

19.5.3 Supply Chain Management

Watsonx.governance contributes to:

- Improved demand forecasting.
- Increased forecasting accuracy.
- Greater process efficiency.
- Higher productivity.

19.5.4 Recruitment

Watsonx.governance supports fair hiring practices by:

- Ensuring unbiased decision-making.
- Detecting and mitigating bias.
- Providing documentation to support auditing efforts.

19.6 Key Takeaways

- **IBM watsonx** consists of three core components:
  - watsonx.ai
  - watsonx.data
  - watsonx.governance
- **watsonx.ai** combines foundation models and traditional machine learning into a comprehensive studio that supports the AI lifecycle.
- **watsonx.data** is a curated data repository built on an open lakehouse architecture. It supports AI model training and fine-tuning while simplifying data interaction through the foundation models of watsonx.ai.
- **watsonx.governance** is a comprehensive AI governance toolkit that enables organizations to direct, manage, monitor, and explain AI systems while ensuring fairness, transparency, compliance, and visibility through customizable dashboards, reports, and charts.

20 Hugging Face

20.1 Learning Objectives

After studying this topic, we will be able to:

- Explain the purpose of the Hugging Face platform.
- List the tools and capabilities offered by Hugging Face.

- Understand how Hugging Face and IBM watsonx.ai jointly help businesses.

20.2 Introduction to Hugging Face

Hugging Face is an open-source Artificial Intelligence (AI) platform where scientists, developers, and businesses collaborate to build personalized machine learning tools.

The platform was created to serve as a central hub for the open-source AI community, enabling users to share:

- Machine learning models
- Datasets
- AI applications

This approach makes AI accessible to a wide range of users, including those who do not have the budget or resources to build machine learning applications independently.

Hugging Face is widely recognized for **democratizing AI**, allowing everyone to benefit from smaller, curated models instead of relying on the assumption that a single model can solve every problem.

20.3 Evolution of the Platform

Initially, the Hugging Face community focused primarily on developing **Transformer-based models** to leverage the capabilities of **Natural Language Processing (NLP)**.

Today, the platform supports a wide variety of machine learning applications, including:

- Text generation
- Image generation
- Audio generation
- Video generation

20.4 Hugging Face Ecosystem

The Hugging Face platform currently hosts:

- Over **250,000 open models**
- More than **50,000 datasets**
- Around **1 million open demos**

These resources continue to grow as the community contributes new models and applications.

20.5 Open-Source Transformer Library

Scientists and developers use Hugging Face to build, train, and deploy AI models.

They have access to the platform's open-source **Transformer Library**, which contains over **25,000 pretrained models** compatible with:

- PyTorch (a deep learning library)
- TensorFlow (a machine learning platform)
- Google JAX (a machine learning framework)

These pretrained models perform a wide range of tasks, including:

- Text generation
- Question answering
- Text summarization
- Automatic speech recognition
- Image segmentation

Users can:

- Search for existing models using filters.
- Share their own pretrained models with the community.

20.6 Hugging Face Spaces

Developers can host demonstrations of their Generative AI applications using the **Spaces** tab.

This allows users to:

- Interact with applications.
- Test their functionality.
- Validate model performance before deployment.

20.7 Benefits for Businesses

Hugging Face provides businesses with an **Enterprise Hub**, which offers access to pretrained models and datasets.

This enables organizations to leverage existing AI infrastructure instead of building models from scratch.

As a result, businesses benefit from:

- Reduced development time
- Lower implementation costs
- Reduced carbon footprint
- Faster scalability

Organizations can also fine-tune pretrained models using proprietary data and business-specific use cases.

20.8 Additional Business Capabilities

Hugging Face helps businesses to:

- Add or remove model features to improve efficiency.
- Evaluate generative AI models to identify and filter biased data.
- Create multimodal applications capable of generating:
    - Text
    - Images
    - Audio
    - Video

20.9 Industry Adoption

More than **50,000 large and small companies** actively use Hugging Face.

20.9.1 Writer

Writer, a Generative AI solution provider, hosts its **Palmyra Large Language Models (LLMs)** on Hugging Face.

20.9.2 Intel

Intel has joined the **Hugging Face Hardware Partner Program** and collaborates with Hugging Face to build:

- State-of-the-art machine learning hardware.
- End-to-end machine learning workflows.

20.9.3 Universities and Non-Profit Organizations

Hugging Face is also widely used by:

- Universities
- Research institutions
- Non-profit organizations

20.10 Additional Services

Beyond models and datasets, Hugging Face offers several additional services.

20.10.1Expert Acceleration Program

This program helps guide non-developers in understanding and working with machine learning models.

20.10.2 HuggingChat

HuggingChat is the first open-source alternative to ChatGPT.

20.11 Security and Compliance

To protect its users, Hugging Face complies with **Service Organization Control (SOC) Type 2** regulations.

This ensures:

- Data security
- Availability
- Processing integrity
- Confidentiality
- Privacy

20.12 Partnership Between Hugging Face and IBM watsonx.ai

Hugging Face has established a strategic partnership with **IBM watsonx.ai**, IBM's next-generation enterprise AI studio.

20.12.1 How watsonx.ai Benefits

IBM watsonx.ai provides selected Hugging Face models within its AI studio, enabling developers to:

- Train machine learning models.
- Test AI solutions.
- Deploy machine learning applications.
- Build Generative AI applications.

This partnership allows IBM watsonx.ai to leverage:

- The diversity of Hugging Face models.
- Community-driven innovation.
- Open-source libraries.

20.12.2 How Hugging Face Benefits

Hugging Face creates open-source versions of IBM's Large Language Models (LLMs) and makes them available through its platform.

Both organizations share a common belief in:

- Open-source AI
- Community-driven innovation
- Collaborative development

20.13 Competitive Advantage

As proprietary AI models can quickly become obsolete, Hugging Face may develop an advantage over major AI companies such as:

- Google
- OpenAI
- Meta
- IBM
- Microsoft

This advantage comes from its strong support for, and contributions from, the open-source AI community, which continuously drives innovation.

20.14 Key Takeaways

- Hugging Face is an open-source AI platform that enables scientists, developers, businesses, universities, and non-profit organizations to collaborate on machine learning projects.
- The platform hosts hundreds of thousands of open models, tens of thousands of datasets, and millions of interactive demos.
- Its Transformer Library provides over 25,000 pretrained models compatible with PyTorch, TensorFlow, and Google JAX.
- Hugging Face Spaces allows developers to share and demonstrate Generative AI applications.
- Businesses benefit from pretrained models, reduced costs, faster deployment, lower carbon footprint, and multimodal AI capabilities.
- Additional offerings include the Expert Acceleration Program, HuggingChat, and SOC Type 2 compliance for security and privacy.
- Through its partnership with IBM watsonx.ai, Hugging Face strengthens enterprise AI development by combining open-source innovation with enterprise-grade AI tools.
- Hugging Face demonstrates that organizations of all sizes—not just large enterprises—can successfully leverage Generative AI.

21 Lesson Summary

At this point, we have learned about the pre-trained models and platforms used for Generative AI application development.

We explored the ability of foundation models to generate text, images, and code using pre-trained models. We learned about the features and capabilities of different platforms, including IBM watsonx and Hugging Face. We also gained hands-on experience with AI application development and IBM watsonx.ai through practical lab exercises. In addition, we explored insights shared by industry experts on AI application development.

21.1 Key Learning Points

- **Text-to-text generation models** are a type of machine learning model used to generate text from a given input.
- There are two types of text-to-text generation models:
    - Statistical models
    - Neural network models
- Text-to-text generation models use **Seq2Seq (Sequence-to-Sequence)** or **Transformer** architectures.
- The most popular text-to-text generation models include:
    - GPT
    - T5
    - BART
- Text-to-text generation models improve the efficiency of tasks such as:
    - Writing scripts
    - Composing musical pieces
    - Drafting emails
    - Writing letters
    - Other text generation tasks
- **Text-to-image generation models** generate images from text descriptions. They interpret the meaning of text prompts and convert them into unique images.
- Text-to-image generation models are classified into two types:
    - Generative Adversarial Networks (GANs)
    - Diffusion Models
- The most popular text-to-image generation models include:
    - DALL·E
    - Imagen
- Text-to-image generation models can generate various types of realistic and abstract images.
- **Text-to-code generation models** are machine learning models that generate source code from natural language descriptions.
- There are two types of text-to-code generation models:
    - Seq2Seq
    - Transformer
- The most popular text-to-code generation models include:
    - CodeT5
    - Code2Seq
    - PanGu-Coder
- Text-to-code generation models support automatic coding, making software development more efficient, accurate, and consistent than manual coding.
- **IBM watsonx** is an AI and data platform that helps organizations build AI responsibly and transparently. It consists of three products:
    - watsonx.ai
    - watsonx.data
    - watsonx.governance
- Together, these three products enable IBM watsonx to amplify the impact of AI across businesses.
- **Hugging Face** is an open-source AI platform where scientists, entrepreneurs, developers, and individuals collaborate to build personalized machine learning tools and models at a reduced cost, within an accelerated timeframe, and with a lower carbon

footprint(Sharing and fine-tuning open-source models reduces computer processing time, cutting down the energy and emissions needed to train AI from scratch).

Module 3: Wrap up

22 Glossary

The following alphabetized glossary contains the names and definitions of specialized terms used throughout the course. Understanding these terms helps us better comprehend the concepts covered in the course.

Term	Definition
Artificial Neural Networks (ANNs)	A collection of smaller computing units called neurons that are modeled in a manner similar to how the human brain processes information.
Bidirectional Autoregressive Transformer Model (BART)	A text-to-text transfer transformer model developed by Facebook AI with a seq2seq translation architecture, featuring a bidirectional encoder representation like BERT and a left-to-right decoder like GPT.
Bidirectional Encoder Representations from Transformers (BERT)	A family of language models developed by Google that uses pre-training and fine-tuning to create models capable of accomplishing several language tasks.
Chatbot	A computer program that simulates human conversation with an end user. Though not all chatbots are equipped with Artificial Intelligence (AI), modern chatbots increasingly use conversational AI techniques such as Natural Language Processing (NLP) to understand users' questions and automate responses.
Clustering	An application of unsupervised learning in which algorithms group similar instances together based on their inherent properties.
Code2Seq	A text-to-code seq2seq model developed by OpenAI and trained on a substantial text and code dataset. It leverages the syntactic structure of programming languages to encode source code.
CodeT5	A text-to-code seq2seq model developed by Google AI and trained on a large dataset of text and code. CodeT5 is the first pre-trained programming language model that is code-aware and encoder-decoder based.
Convolutional Neural Networks (CNNs)	Deep learning architecture networks containing a series of layers, each performing a convolution (mathematical operation) on the previous layer.
DALL·E	A text-to-image generation model developed by OpenAI that is trained on a large dataset of text and images and generates realistic images from text descriptions.
Deep Learning	A type of machine learning focused on training computers to perform tasks through learning from data. It uses artificial neural networks.

Diffusion Model	A type of generative model commonly used for generating high-quality samples and performing tasks such as image synthesis. It is trained by gradually adding noise to an image and then learning to remove that noise through a process called diffusion.
Dimensionality Reduction	An application of unsupervised learning in which algorithms capture the most essential data features while discarding redundant or less informative ones.
Falcon	A large language model developed by the Technology Innovation Institute (TII). Its variant, Falcon-7B-Instruct, is a 7-billion-parameter decoder-only model.
Foundational Models	AI models with broad capabilities that can be adapted to create more specialized models or tools for specific use cases.
Generative Adversarial Network (GAN)	A type of generative model that includes two neural networks: a generator and a discriminator. The generator creates samples such as text and images, while the discriminator determines whether the generated samples are real or fake.
Generative AI Models	Models that understand the context of input content to generate new content. They are generally used for automated content creation and interactive communication.
Generative Pre-trained Transformer (GPT)	A series of large language models developed by OpenAI that understand language by combining pre-training and transformer architectures.
Google Flan	An encoder-decoder foundational model based on the T5 architecture.
Google JAX	A machine learning framework used for transforming numerical functions. It combines autograd (automatic gradient computation through differentiation) with TensorFlow's XLA (Accelerated Linear Algebra).
Hugging Face	An AI platform that enables open-source researchers, entrepreneurs, developers, and individuals to collaborate and build personalized machine learning tools and models.
IBM Granite	Multi-size foundational models specifically designed for businesses. These models use a decoder architecture to apply Generative AI to both language and code.
IBM watsonx	An integrated AI and data platform with AI assistants designed to scale and accelerate AI adoption using trusted data across businesses.
Imagen	A text-to-image generation model developed by Google AI that generates realistic images from text descriptions.
Large Language Models (LLMs)	Deep learning models trained on substantial text datasets to learn language patterns and structures. They perform tasks such as text generation, translation, summarization, sentiment analysis, and more.
Llama	A large language model developed by Meta AI.
Natural Language Processing (NLP)	A subset of Artificial Intelligence that enables computers to understand, manipulate, and generate human language.

Neural Code Generation	A process that uses artificial neural networks to generate code from natural language inputs or other programming-related prompts.
Neural Network Model	A type of text-to-text generation model that uses artificial neural networks to generate text.
Neural Networks	Computational models inspired by the structure and functioning of the human brain. They are a fundamental component of deep learning and Artificial Intelligence.
Open Lakehouse Architecture	A data lakehouse architecture that combines the features of data lakes and data warehouses.
PanGu-Coder	A text-to-code transformer model developed by Microsoft Research. It is a pre-trained decoder-only language model that generates code from natural language descriptions.
Pre-trained Models	Machine learning models trained on extensive datasets before being fine-tuned for specific tasks or applications. They represent a form of transfer learning where knowledge gained during pre-training is applied to a different task during fine-tuning.
Pre-training	A technique in which unsupervised algorithms are repeatedly allowed to discover relationships between diverse pieces of information.
Prompt	An instruction or question given to a Generative AI model to generate new content.
PyTorch	An open-source machine learning framework based on the Torch library, commonly used for computer vision and Natural Language Processing applications.
Recurrent Neural Networks (RNNs)	A deep learning architecture designed to process sequences of data by maintaining hidden states that capture information from previous sequence steps.
Seq2Seq Model	A text-to-text generation model that first encodes input text into a sequence of numbers and then decodes it into a new sequence representing the generated text.
Statistical Model	A type of text-to-text generation model that uses statistical techniques to generate text.
Supervised Learning	A subset of AI and machine learning that uses labeled datasets to train algorithms to classify data or accurately predict outcomes.
T5	A text-to-text transfer transformer model developed by Google AI and trained on a substantial dataset of code and text. It supports tasks such as summarization, translation, and question answering.
TensorFlow	A free and open-source software library used for machine learning and Artificial Intelligence.
Text-to-Code Generation Model	A machine learning model that generates code from natural language descriptions. It uses Generative AI through neural code generation techniques.
Text-to-Image Generation Model	A machine learning model that generates images from text descriptions by interpreting the meaning of words and converting them into unique images.

Text-to-Text Generation Model	A machine learning model trained on large text corpora to learn language patterns, grammar, and causal relationships. Given an input, it generates new text.
Training Data	Data, generally consisting of large datasets with examples, used to teach a machine learning model.
Transformers	A deep learning architecture that uses an encoder-decoder mechanism to generate coherent and contextually relevant text.
Unsupervised Learning	A subset of machine learning and Artificial Intelligence that uses algorithms to analyze and cluster unlabeled datasets, discovering hidden patterns or groupings without human intervention.
Variational Autoencoder (VAE)	A generative neural network model that learns an efficient representation of input data by encoding it into a smaller latent space and decoding it back to its original form.
watsonx.ai	A studio of integrated tools for working with Generative AI capabilities powered by foundational models and for building machine learning models.
watsonx.data	A massive, curated data repository that can be used to train and fine-tune models using a state-of-the-art data management system.
watsonx.governance	A toolkit used to direct, manage, and monitor an organization's Artificial Intelligence activities.

23 Working with IBM Granite Foundation Models

23.1 Introduction

IBM Granite is a family of enterprise-grade foundation models developed by IBM and trained on enterprise-relevant datasets from domains such as the Internet, academia, software development, legal, and finance.

The **granite-8b-code-instruct** model supports tasks including text generation, summarization, question answering, classification, information extraction, and code generation, making it suitable for a wide range of enterprise AI applications.

23.2 Learning Objectives

After completing these exercises, we are able to:

- Identify the primary use cases of IBM Granite foundation models.
- Explore text summarization using **granite-8b-code-instruct**.
- Utilize IBM Granite models for context-aware question answering.
- Understand how enterprise foundation models can support domain-specific applications, including finance-related tasks.

23.3 Exercise 1: Code Generation Using IBM Granite

This exercise demonstrates how **granite-8b-code-instruct** can generate source code from natural language prompts, enabling AI-assisted software development and programming support.

23.3.1 Foundation Model

- **granite-8b-code-instruct**

23.3.2 Prompt Template

System Instruction

Act as a coding expert and generate efficient, well-structured, and well-documented code that solves the specified programming task.

Example User Prompts

How can we write a Python program to calculate the factorial of a given number?
How can we reverse a string in Python without using built-in functions?
How can we write a Python program to count the number of vowels in a given string?

23.3.3 Practical Applications

Using appropriate prompts, the model can:

- Generate source code in multiple programming languages.
- Solve algorithmic and programming problems.
- Produce readable code with explanations and comments.
- Assist us in learning programming concepts.
- Accelerate software development through AI-assisted coding.

23.4 Exercise 2: Context-Aware Question Answering

This exercise demonstrates how **granite-8b-code-instruct** can answer questions using information from a provided document or knowledge source. By supplying contextual information through prompt instructions, the model generates responses grounded in the given content rather than relying solely on its pre-trained knowledge.

23.4.1 Foundation Model

- **granite-8b-code-instruct**

23.4.2 Prompt Template

System Instruction

Answer all questions using only the information provided in the given article or document. If the answer is not available in the provided content, respond with "I don't know."

Example User Prompts

Question: Is growing tomatoes easy?
Question: What varieties of tomatoes are there?
Question: Why should we use mulch when growing tomatoes?

23.4.3 Practical Applications

This capability can be used to:

- Build document-based question-answering systems.
- Develop intelligent knowledge assistants.
- Create AI-powered customer support systems.
- Extract information from reports, manuals, and documentation.
- Improve enterprise search and knowledge retrieval.

23.5 Exercise 3: Finance-Related Question Answering

This exercise demonstrates how **granite-8b-code-instruct** can answer finance-related questions using its enterprise knowledge. Since IBM Granite models are trained on enterprise-relevant domains, they can assist with understanding financial concepts and answering domain-specific queries.

23.5.1 Foundation Model

- **granite-8b-code-instruct**

23.5.2 Example User Prompts

Define interest rates.
How are interest rates decided?
How do interest rates affect savings accounts?

23.5.3 Practical Applications

The model can assist with:

- Explaining financial concepts.
- Answering finance-related questions.
- Supporting financial education.
- Developing AI-powered financial assistants.
- Creating enterprise knowledge applications for domain-specific information retrieval.

23.6 Summary

By completing these exercises, we explored how **IBM Granite foundation models** can support enterprise AI applications through code generation, context-aware question answering, and domain-specific knowledge retrieval.

Using effective prompt engineering, Granite models can:

- Generate source code.

- Answer questions based on supplied documents.
- Provide accurate responses across specialized domains such as finance.

These capabilities enable the development of:

- AI-powered coding assistants.
- Document-based question-answering systems.
- Enterprise knowledge assistants.
- Customer support solutions.
- Domain-specific AI applications.

Their enterprise-focused training makes IBM Granite models particularly well-suited for building reliable AI solutions in professional and business environments.